

Announced Rescue Pricing: Strategic Waiting, Screening History, and Market Thickness

(Authors' names blinded for peer review)

A platform that commits to a higher payment after an unfilled request expands continuation supply but also gives incumbent drivers a reason to wait. We study this trade-off when expected market thickness is public, realized local supply is latent, and universal rejection both triggers rescue and screens the continuation pool. We do not impose cutoff behavior. With continuous full-support rider-value and driver-cost distributions, every anonymous symmetric independent WPBE is almost everywhere equivalent to a pure driver cutoff, and the cutoff-equilibrium correspondence is nonempty and compact. Under smooth distributions and no fresh entry, any active flat price admits a sufficiently small rescue increase that strictly raises completion on every nearby equilibrium branch. Convexity of the driver-cost CDF strengthens this result: every active strict menu beats the flat policy with the same first payment for every equilibrium cutoff, whereas a concave CDF can reverse the comparison. In the uniform no-entry core, every menu has a unique cutoff, the optimal second payment has a Lambert- W representation, and global menu design reduces to one dimension. The optimized rescue gain is continuous in market thickness, vanishes in very thin and very thick markets, and attains a positive maximum at a finite thickness; neither strict single-peakedness nor uniqueness of that thickness is asserted. An extension with surviving incumbents and fresh entrants identifies an activity condition for local rescue and shows that supply composition affects strategic waiting even at fixed continuation capacity.

Key words: dynamic pricing, rescue payments, strategic waiting, market thickness, screening, Poisson games

1. Introduction

Recent reporting makes the operational problem concrete. In January 2026, DoorDash warned that severe-weather demand could outstrip available couriers, producing long waits and cancelled orders; its stated response included a temporary fee passed through to active couriers as Peak Pay (DoorDash 2026). Reuters documented a sharper rejection mechanism in Kenya: some ride-hailing drivers asked riders to pay more than the app price and cancelled the request when the rider declined, sending the rider back to search for another driver (Okoth 2024). Continuation supply can change as well. Reuters reported that JD.com doubled its three-month recruitment target for full-time delivery riders from 50,000 to 100,000 (Reuters 2025). These episodes establish the separate forces of scarcity, rejection, higher compensation, rematching, and fresh entry; they do not establish the exact same-request contract studied here. We ask whether committing ex ante to a higher payment after universal rejection rescues a request or instead teaches incumbents to wait for the next offer.

Many service requests are initially unattractive to available suppliers but become urgent after a failed response window. A platform can address this problem by announcing an initial payment and a higher, failure-contingent rescue payment. Announcement makes the policy credible, but it also changes the first response: a supplier may reject today in order to preserve the option of receiving more tomorrow. The platform must therefore design the rescue schedule and the induced waiting equilibrium jointly.

This tension is especially sharp when the platform observes an expected market state but not the realized local supply attached to a particular request. Universal rejection is then both an operational event and an informational event. It triggers the second response window and reveals that every incumbent who was willing to accept the initial payment is absent from the continuation pool. Incumbents who remain are selected by their earlier rejection, whereas drivers who enter later have not been screened. The value of future supply consequently depends on its screening history, not only on its expected quantity.

We study a two-period request in which a platform commits to payments (p_1, p_2) with $p_2 \geq p_1$. A private-value rider first decides whether to post. Private-cost incumbent drivers then simultaneously accept p_1 or wait. After universal rejection, the rider can abandon, repeat p_1 , or activate p_2 . Incumbents may survive to the second window and a fresh pool may enter. The expected incumbent thickness is public, while realized local supply and rival costs remain latent. Drivers who wait discount their terminal payoff, and accepting drivers compete for assignment.

The analysis follows a policy–response–optimization sequence. We first fix the announced menu and characterize weak perfect Bayesian equilibrium among anonymous symmetric independent driver strategies. Cutoff behavior is a result rather than a restriction: the accept-versus-wait payoff difference is strictly decreasing in driver cost, so every equilibrium strategy is almost everywhere a pure cutoff. Universal rejection generates a tractable failure posterior. Incumbents remaining above the cutoff are screened, whereas fresh entrants retain the original cost distribution. This fixed-menu analysis allows general continuous full-support cost and value distributions and makes the rider’s abandon–repeat–rescue decision part of drivers’ continuation payoffs.

The global design problem is exactly solvable when driver costs and rider values are uniform and fresh entry is absent. Every announced menu then has a unique equilibrium cutoff. For a fixed p_1 , the optimal rescue problem separates into reject-all, tangent-rescue, and repeat-only regimes. The optimal continuation price has a Lambert- W representation, and joint design of (p_1, p_2) reduces to a continuous one-dimensional problem. Above an endogenous rider-patience threshold, the optimized rescue menu strictly improves completion relative to the optimized flat payment. Positive incumbent survival makes $\beta \geq 1/2$ a simple sufficient condition at every finite thickness.

The uniform distribution is used for this exact global solution, not for the cutoff logic. Under general smooth distributions and no entry, a sufficiently small rescue increase at any active flat price raises completion on every nearby equilibrium branch. Under a convex driver-cost CDF, every active strict menu improves on the flat policy with the same p_1 for every equilibrium cutoff. A concave-CDF example reverses the comparison, and a continuous full-support example has three equilibrium cutoffs. These results identify which conclusions survive beyond the exact core and why equilibrium selection cannot be ignored in unrestricted distributions.

We next return to the general continuation pool. When $\alpha < 1$ and $\gamma > 0$, the second window contains both screened surviving incumbents and unscreened new entrants. A local theorem shows when a small announced rescue increase improves completion after accounting for both sources of supply. If at least one optimized flat price satisfies this activity condition, the same perturbation proves strict policy-class dominance over optimized flat pricing, without requiring the flat optimizer to be unique. A separate composition theorem holds either the expected continuation headcount or the local price-marginal capacity fixed. Replacing entrants with screened incumbents strictly strengthens first-period acceptance erosion; its effect on the local completion gain has no unconditional sign and can reverse with δ .

Market thickness determines where rescue design matters. Let $M_R^*(m)$ and $M_F^*(m)$ denote the optimized completion probabilities under rescue and flat pricing at the same thickness. Their difference $V(m)$ is continuous, vanishes in very thin and very thick markets, and is strictly positive at some finite thickness. Hence a finite intermediate market maximizes the rescue gain. This existence result does not imply that $V(m)$ is strictly single-peaked or that the maximizing thickness is unique.

This paper makes three contributions. First, it characterizes the endogenous cutoff-WPBE generated jointly by ex ante menu commitment, latent local competition, universal-rejection updating, and private rider continuation. General-distribution existence and purification separate this equilibrium foundation from the uniform formulas used later. Second, it solves the uniform no-entry menu-design problem globally and establishes how rider patience and market thickness govern the incremental completion value of rescue pricing. Third, it maps the boundary of those global results: smooth and convex distributional extensions identify robust rescue gains, while concavity and multiplicity examples delimit them; the mixed-supply extension separates screened surviving incumbents from unscreened fresh entrants and shows why continuation composition matters. The contribution is this joint equilibrium-design structure, not two-period pricing, Poisson supply, strategic waiting, or entry in isolation.

The remainder of the paper reviews the closest literatures, develops the institution and general fixed-menu equilibrium, establishes distributional robustness and its limits, solves the exact uniform

design problem, studies market thickness and continuation-supply composition, and closes with institutional scope. Proofs and numerical verification details appear in the appendices.

2. Literature Review

Procurement clocks and strategic timing. [Li and Kuo \(2013\)](#) optimize a finite discrete Dutch auction that advances after universal nonacceptance, permits a Poisson number of bidders, and randomly selects among simultaneous acceptors. Reversing the transfer direction gives the low-to-high procurement skeleton of an announced rescue menu. [Buchanan et al. \(2016\)](#) derive symmetric Bayesian bidding when realized group size is hidden and nonacceptance changes beliefs along a multi-unit Dutch clock. [Carare and Rothkopf \(2005\)](#) study slow Dutch auctions with waiting costs, and [Shneyerov \(2014\)](#) optimizes a dynamic clock when market sides discount at different rates. These papers establish the importance of strategic timing, population uncertainty, and learning from nonacceptance. Our posted-payment environment instead embeds the clock in a private rider's continuation problem and evaluates unconditional request completion.

Search mechanisms provide a second foundation for the continuation pool. [McAfee and McMillan \(1988\)](#) study dynamic mechanisms that acquire additional participants over time. [Cr  mer et al. \(2007\)](#) formulate optimal search auctions in which previously contacted bidders are incumbents and later search brings new entrants; a rejected incumbent can therefore remain in a subsequent allocation with a newly solicited bidder. These papers establish retained incumbency and fresh entry before strategic delay is added.

[Lee and Li \(2023\)](#) is especially close to the survival-and-entry extension. Their seller commits to a reserve-price and sampling schedule; long-lived incumbent bidders can bid or wait, new bidders enter later, and earlier screening truncates the values of the incumbent pool while new bidders retain the original distribution. Under the procurement dual, this is the same screened-incumbent versus unscreened-entrant distinction that appears here. Their analysis establishes announced price paths, strategic waiting, and the mixed composition of future supply in a search-auction environment. Our setting instead uses an anonymous posted-payment allocation with latent Poisson competition, a rider who chooses among abandon, repeat, and rescue, and completion comparisons between optimized rescue and flat policies.

Announced dynamic pricing and forward-looking agents. [Aviv and Pazgal \(2008\)](#) solve announced markdown design by fixing a price path, deriving a forward-looking consumer threshold, and then optimizing the path. [Correa et al. \(2016\)](#) study contingent pricing with strategic consumers. These papers supply the methodological policy–response–optimization order, but their consumers wait for a lower price and face stockout. Here suppliers wait for a higher payment and face assignment competition. In two-sided markets, [Chen and Hu \(2020\)](#) study Poisson arrivals with strategic

waiting, and [Hu et al. \(2022\)](#) obtain both high–low and low–high surge paths from rider and driver temporal responses. The present model isolates a single request, an ex ante failure-contingent menu, and the posterior created by universal rejection.

Platform bonuses, dispatch, and strategic rejection. [Wu et al. \(2022\)](#) optimize multistage order-specific bonuses in meal delivery using an acceptance model driven by the current bonus. Their fixed-later-bonus ablation finds no detectable anticipation before that bonus, although the future path is not described as disclosed and committed. This evidence is an important institutional contrast: our strategic channel requires the complete failure-contingent menu to be credible before first-window decisions. [Sigg et al. \(2025\)](#) model workers who reject a delivery request so that it is reoffered at a higher payment. Broadcast, supplier-menu, and notification models also study contention and a second attempt after no response, but vary outreach, assignment, or demand batching rather than solve the same-request announced payment game ([Sun et al. 2020](#), [Horner et al. 2021](#), [Ausseil et al. 2022](#), [Feng et al. 2023](#), [Qin et al. 2025](#)).

Latent supply, strategic availability, and market thickness. [Chen \(2012\)](#) derives private-cost sellers' willingness to wait for higher buyer offers when the seller count is fixed and public and the buyer does not precommit. [Bai et al. \(2023\)](#) let a platform announce a supply-contingent price and compensation rule when true supply is random and providers can hide availability. Strategic rejection, relocation, and acceptance are likewise established in ride-hailing and service-platform models ([Guda and Subramanian 2019](#), [Garg and Nazerzadeh 2022](#), [Afèche et al. 2023](#), [Feng and Wang 2023](#), [Meskar et al. 2023](#)). Poisson population uncertainty determines our assignment and failure posteriors, but Poisson supply and market thickness are tools rather than standalone contributions.

Market thickness itself has direct theoretical and operational precedents. [Loertscher et al. \(2022\)](#) balance gains from accumulating traders against delay in a dynamic bilateral-trade mechanism. Using online food-delivery operations, [Zhao et al. \(2024\)](#) show how processing time changes the thickness available for matching. Together they establish the broader economics of intermediate and optimally chosen market thickness. Our object holds m fixed and compares two payment-policy classes: $V(m)$ is the completion advantage of optimal rescue pricing over optimal flat pricing in that same environment.

Population uncertainty and monotone behavior. [Myerson \(1998\)](#) develops Poisson games as a tractable representation of population uncertainty, while [Athey \(2001\)](#) gives general conditions for monotone pure strategies in games with one-dimensional private information. Our cutoff result is model-specific and does not follow mechanically from either foundation: after failure, a driver's continuation payoff depends on the rider's endogenous choice among abandon, repeat, and rescue. The affine accept-versus-wait margin lets us prove directly that anonymous symmetric independent equilibrium behavior is almost everywhere a pure cutoff.

Table 1 Model primitives.

Symbol	Domain	Economic meaning
p_1, p_2	$0 \leq p_1 \leq p_2 \leq 1$	Initial payment and committed payment after universal rejection
m	$(0, \infty)$	Public expected thickness of the incumbent pool
N^I	$\text{Pois}(m)$	Latent realized incumbent count
α	$[0, 1]$	Probability that a rejected incumbent remains available
γ	$[0, \infty)$	Fresh-entry intensity relative to m
β	$(0, 1)$	Rider continuation factor after delay
δ	$(0, 1]$	Incumbent payoff discount from waiting
F	CDF on $[0, 1]$	Incumbent and entrant cost distribution
G	CDF on $[0, 1]$	Rider value distribution

The parameters α and δ govern different margins: physical availability and intertemporal incentives. Expected thickness m is distinct from the latent realization N^I .

Against these literatures, we characterize the endogenous cutoff-WPBE and completion-design problem created jointly by an ex ante same-request posted-payment menu, latent realized competition, a private rider's post-failure continuation, and a continuation pool containing screened incumbents and potentially unscreened entrants. The global design and thickness theorems are proved for the stated uniform no-entry core; the general-distribution and entry results state precisely which conclusions survive outside that core.

3. Model

3.1. Primitives and policy

Table 1 defines the primitives once. The fixed-menu equilibrium analysis allows general continuous, strictly increasing, full-support CDFs F and G on $[0, 1]$. Uniform distributions are imposed only where stated for the exact global design and thickness analysis.

There is one potential request and two periods. Market thickness $m > 0$ is public. Before rider or driver private information is used, the platform commits to

$$\mathcal{P}^D = \{(p_1, p_2) : 0 \leq p_1 \leq p_2 \leq 1\}.$$

The second payment is available only following first-period failure and cannot be revised ex post. A flat policy is (p, p) . Payment is a transfer from the rider to the selected driver. The platform maximizes the unconditional probability that the potential request is completed; it does not maximize profit or welfare.

The rider has value $v \sim G$. A period-1 completion yields $v - p_1$ and a period-2 completion at payment p_j yields $\beta v - p_j$, where $\beta \in (0, 1)$. Incumbents have a separate discount factor $\delta \in (0, 1]$: if an incumbent waits and is selected in period 2, her period-1-equivalent net payoff is $\delta(p_j - c)$. A fresh entrant arrives only in period 2 and evaluates the contemporaneous payoff $p_j - c$. Thus β measures

rider patience and δ measures incumbent-driver patience; the common-discount specification is the restriction $\delta = \beta$. From the request perspective the incumbent count is

$$N^I \sim \text{Pois}(m).$$

The realization is latent to every player. Incumbent costs are i.i.d. F and remain fixed across periods. Throughout the general analysis, F and G are continuous, strictly increasing, and have full support on $[0, 1]$. An accepting driver incurs her cost only if selected; an unselected acceptor earns and incurs zero.

Conditional on the rider continuing after universal rejection, each incumbent survives independently with probability $\alpha \in [0, 1]$ and a fresh pool

$$N^E \sim \text{Pois}(\gamma m), \quad \gamma \geq 0,$$

arrives independently with i.i.d. costs distributed according to F . These realizations occur after the rider's abandon-repeat-rescue decision, and the rider observes neither count nor survival. Moving the independent latent draws immediately before that decision would be outcome-equivalent. At the terminal node, an available driver accepts the active payment p_j exactly when $c \leq p_j$.

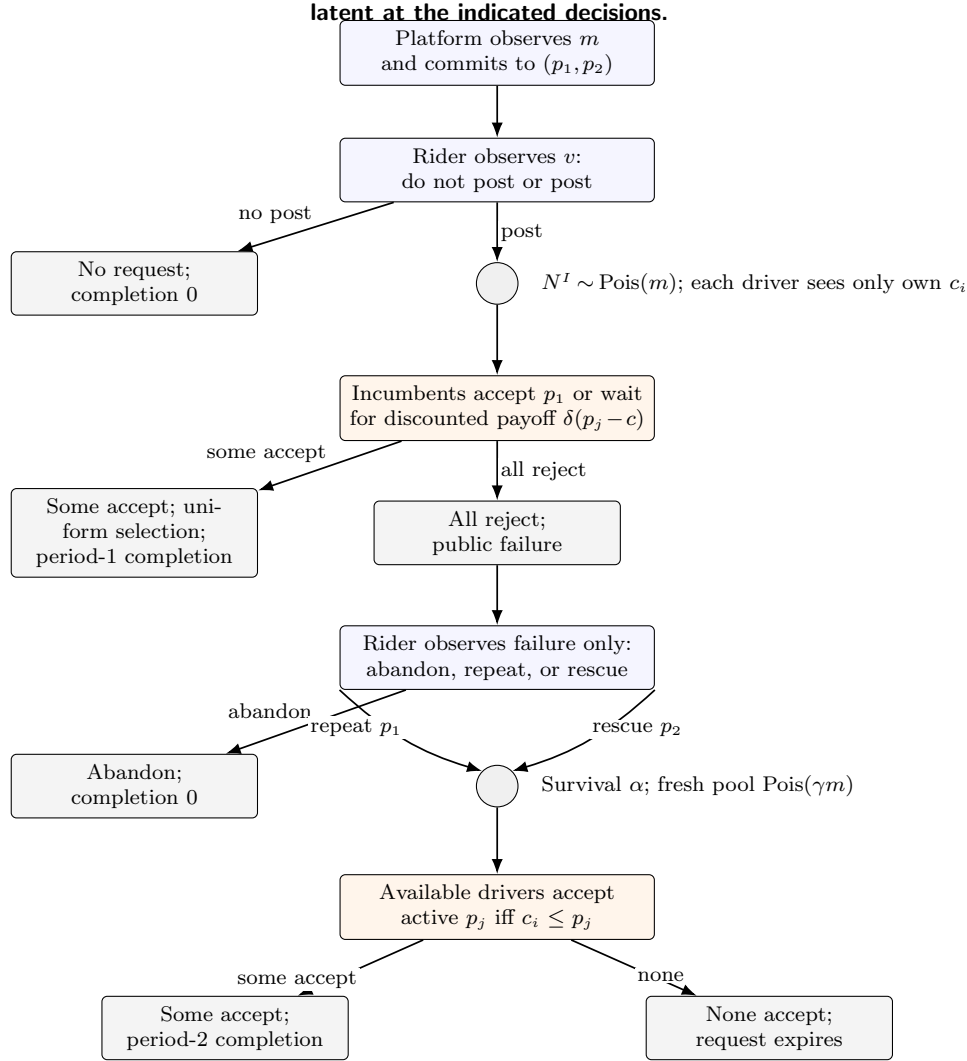
3.2. Timing and information

Figure 1 gives the reduced extensive form. The driver nodes represent simultaneous Bayesian moves, not observation of a realized pool.

The timing is as follows.

- Stage 1.* The platform observes the public state m and commits to the entire menu (p_1, p_2) . The rescue payment cannot be revised after observing failure.
- Stage 2.* The rider privately observes v and chooses whether to post the request. A nonposted request yields zero completion.
- Stage 3.* Conditional on posting, the latent incumbent pool is realized. Each incumbent observes the menu and own cost but not the number or costs of rivals.
- Stage 4.* Incumbents simultaneously accept p_1 or wait. If at least one accepts, one acceptor is selected uniformly and the request completes in period 1.
- Stage 5.* If every incumbent rejects, universal rejection becomes public. It reveals the first-period screening event but not the realized pool size.
- Stage 6.* After observing failure, the rider chooses abandon, repeat p_1 , or activate p_2 .
- Stage 7.* Each rejected incumbent independently survives with probability α . Independently, a fresh unscreened pool with mean γm enters. Counts, survival realizations, and entrant costs remain latent.

Figure 1 Reduced extensive form. The realized incumbent count, rival costs, survival, and fresh entry remain latent at the indicated decisions.



Stage 8. Available drivers accept the active terminal payment when it covers their cost. If at least one accepts, uniform selection produces period-2 completion; otherwise the request expires.

The pair (p_1, p_2) is not an arbitrary truncation of a richer deterministic policy within this institution. It is the complete anonymous transfer menu generated by the two response windows and the rider's publicly observed repeat–rescue choice.

PROPOSITION 1 (Institutional representation). *Suppose the platform commits before types and supply are realized, drivers and riders send no private reports, acceptance ends the request, universal rejection is the only nonterminal public history, and every eligible driver in a window faces the same common rider-to-driver transfer. After rejection the rider publicly chooses abandon, repeat, or rescue; repeat preserves the initial quote and rescue activates one committed alternative*

quote. Every deterministic anonymous transfer schedule in this interface is represented by a pair (p_1, p_2) . If rescue payments cannot fall below the initial quote, the feasible set is $p_1 \leq p_2$.

The initial response and a repeat use p_1 ; the rider's publicly observed rescue action activates p_2 . Proposition 1 is completeness within this action-contingent interface, not optimality among arbitrary direct mechanisms. Reports, more windows, driver-specific offers, or separate rider charges and driver wages create additional instruments.

3.3. Equilibrium class, beliefs, and tie-breaking

We use weak perfect Bayesian equilibrium (WPBE) and initially restrict only first-window driver behavior to anonymous symmetric independent strategies. A driver may use any measurable acceptance probability $\sigma : [0, 1] \rightarrow [0, 1]$. Cutoff behavior will be derived from incentives in Section 4, rather than imposed here. We do not claim to rule out asymmetric or correlated equilibria.

Whenever posting yields zero, the rider posts if and only if $v \geq p_1$. After failure, the following priority rule makes continuation single-valued. If the maximal continuation payoff is zero, the rider repeats iff $\beta v - p_1 \geq 0$ and otherwise abandons. If repeat and rescue tie at a strictly positive maximum, the rider chooses rescue. Terminal drivers accept at a zero surplus. These rules matter when repeat has zero coverage: a positive interval of rider types can then be indifferent between repeat and abandon.

For $p_1 < 1$, posting truncates G to $[p_1, 1]$. Since supply is independent of value, failure leaves this value posterior unchanged. Poisson splitting determines the rider's supply posterior after failure. For a focal incumbent, Palm conditioning leaves the number of rivals distributed as $\text{Pois}(m)$. At $(p_1, p_2) = (1, 1)$, posting is a null event; we complete the off-path driver subgame with cutoff one and define completion as zero.

4. Equilibrium under an Arbitrary Announced Menu

Fix a candidate cutoff $a \in [0, p_1]$. Extend the rider CDF outside its support by

$$\tilde{G}(x) = \begin{cases} 0, & x \leq 0, \\ G(x), & 0 < x < 1, \\ 1, & x \geq 1, \end{cases} \quad \phi(x) = \begin{cases} (1 - e^{-x})/x, & x > 0, \\ 1, & x = 0. \end{cases} \quad (1)$$

The terminal eligible intensity and coverage at payment p_j are

$$\lambda_j(a) = m\{\alpha[F(p_j) - F(a)] + \gamma F(p_j)\}, \quad C_j(a) = 1 - e^{-\lambda_j(a)}. \quad (2)$$

LEMMA 1 (Poisson competition and failure posterior). *Under cutoff a , first-window completion conditional on posting is $1 - e^{-mF(a)}$. A focal acceptor's expected assignment share is $\phi(mF(a))$, while a focal waiting driver reaches failure against all rivals with probability $e^{-mF(a)}$.*

Conditional on universal rejection, incumbents above a remain a Poisson marked population. After survival and fresh entry, terminal coverage at p_j is $C_j(a)$, and an eligible focal driver's terminal assignment share is $\phi(\lambda_j(a))$.

The assignment identity is

$$\mathbb{E} \left[\frac{1}{1+Z} \right] = \int_0^1 \mathbb{E}[t^Z] dt = \frac{1-e^{-x}}{x}, \quad Z \sim \text{Pois}(x).$$

Poisson marking makes counts in disjoint cost regions independent (Myerson 1998). Conditional on no cost below a , the process above a is unchanged; it is not the law of $N^I - 1$ conditional on $N^I \geq 1$.

After failure, rider payoffs from abandon, repeat, and rescue are

$$0, \quad C_1(a)(\beta v - p_1), \quad C_2(a)(\beta v - p_2).$$

If $C_2(a) > C_1(a)$, define

$$v^M(a) = \frac{C_2(a)p_2 - C_1(a)p_1}{\beta[C_2(a) - C_1(a)]};$$

otherwise put $v^M(a) = +\infty$. Conditional on posting, the repeat and rescue masses are

$$\eta^R(a) = \frac{\tilde{G}(v^M(a)) - \tilde{G}(p_1/\beta)}{1 - G(p_1)}, \quad \eta^2(a) = \frac{1 - \tilde{G}(v^M(a))}{1 - G(p_1)}, \quad (3)$$

and

$$\eta^R(a) + \eta^2(a) = \rho_G(p_1) := \frac{1 - \tilde{G}(p_1/\beta)}{1 - G(p_1)}. \quad (4)$$

The switch satisfies $v^M(a) \geq p_1/\beta$, so the masses in (3) are nonnegative. The initial posting payoff is

$$(1 - e^{-mF(a)})(v - p_1) + e^{-mF(a)} \max\{0, C_1(\beta v - p_1), C_2(\beta v - p_2)\}.$$

Together with the tie-breaking rule, this implies posting exactly when $v \geq p_1$.

We next show that the cutoff is generated by equilibrium incentives. Let $\sigma(c)$ be an arbitrary anonymous symmetric independent first-window acceptance probability and put $q = \int \sigma(c) dF(c)$.

The rival terminal intensity is

$$\lambda_j(\sigma) = m \left\{ \alpha \int_0^{p_j} [1 - \sigma(t)] dF(t) + \gamma F(p_j) \right\}. \quad (5)$$

The rider masses are defined as in (3), using the coverages induced by σ .

THEOREM 1 (Endogenous cutoff). *Fix $0 < p_1 < 1$ and $p_1 \leq p_2 \leq 1$. In every anonymous symmetric WPBE with independent driver mixing, first-window behavior is F -almost everywhere equivalent to a pure cutoff: for some $a \in [0, p_1]$, costs below a accept and costs above a wait. Mixing can occur only at the null type $c = a$. Conditional on any public sunspot realization the same conclusion holds; a sunspot can mix across numerical cutoffs only when the underlying cutoff equilibrium is multiple.*

For $c \leq p_1$, accepting and waiting yield

$$A(c; \sigma) = \phi(mq)(p_1 - c),$$

$$W(c; \sigma) = \delta\alpha e^{-mq} \{ \eta^R \phi(\lambda_1)(p_1 - c) + \eta^2 \phi(\lambda_2)(p_2 - c) \}.$$

Their difference is

$$A(c; \sigma) - W(c; \sigma) = B(\sigma)(p_1 - c) - D(\sigma), \quad (6)$$

where

$$B(\sigma) = \phi(mq) - \delta\alpha e^{-mq} \{ \eta^R \phi(\lambda_1) + \eta^2 \phi(\lambda_2) \},$$

$$D(\sigma) = \delta\alpha e^{-mq} \eta^2 \phi(\lambda_2)(p_2 - p_1).$$

If $q > 0$, then $B(\sigma) \geq \phi(mq) - e^{-mq} > 0$. If $q = 0$, then $B(\sigma) \geq 1 - \delta\alpha\rho_G(p_1) > 0$. Thus (6) is strictly decreasing in cost and crosses zero at most once. Theorem 1 is therefore a direct implication of this model's payoff geometry, not an assumed monotonicity restriction.

For a cutoff a , define the cutoff-type margin

$$f_{F,G}(a; p_1, p_2) = \phi(mF(a))(p_1 - a) - \delta\alpha e^{-mF(a)} \{ \eta^R(a) \phi(\lambda_1(a))(p_1 - a) + \eta^2(a) \phi(\lambda_2(a))(p_2 - a) \}. \quad (7)$$

THEOREM 2 (Cutoff-WPBE existence and characterization). *For $0 < p_1 < 1$, the cutoff equilibrium set is*

$$\begin{aligned} \mathcal{E}_{F,G}(p_1, p_2) = & \{0 : f_{F,G}(0; p_1, p_2) \leq 0\} \\ & \cup \{a \in (0, p_1) : f_{F,G}(a; p_1, p_2) = 0\} \\ & \cup \{p_1 : f_{F,G}(p_1; p_1, p_2) \geq 0\}. \end{aligned} \quad (8)$$

It is nonempty and compact. At $p_1 = 0$, the only aggregate cutoff is zero with the lower-boundary reject action; at $(1, 1)$, set $\mathcal{E}_{F,G}(1, 1) = \{1\}$. Hence every feasible menu admits an anonymous symmetric WPBE, and Theorem 1 represents every anonymous symmetric independently mixed WPBE by an element of this set.

At the upper boundary,

$$f_{F,G}(p_1; p_1, p_2) = -\delta\alpha e^{-mF(p_1)} \eta^2(p_1) \phi(\lambda_2(p_1))(p_2 - p_1) \leq 0.$$

Continuity, the lower-boundary test, and the intermediate value theorem give existence; closedness gives compactness. General distributions do not imply that the set is single valued.

Completion on cutoff branch a is

$$M_{F,G}^{\alpha,\gamma}(p_1, p_2; a) = [1 - G(p_1)] [1 - e^{-mF(a)} + e^{-mF(a)} \{ \eta^R(a) C_1(a) + \eta^2(a) C_2(a) \}]. \quad (9)$$

The conservative fixed-menu value is

$$\underline{M}_{F,G}^{\alpha,\delta,\gamma}(p_1, p_2) = \min_{a \in \mathcal{E}_{F,G}(p_1, p_2)} M_{F,G}^{\alpha,\gamma}(p_1, p_2; a), \quad (10)$$

and the rescue-policy value is

$$M_R^*(m; F, G, \alpha, \beta, \delta, \gamma) = \sup_{(p_1, p_2) \in \mathcal{P}^D} \underline{M}_{F, G}^{\alpha, \delta, \gamma}(p_1, p_2). \quad (11)$$

The minimum is well defined. We retain a supremum for unrestricted distributions; later sections prove attainment for the exact uniform no-entry core.

5. Flat-Payment Benchmark

PROPOSITION 2 (Flat policy under general distributions). *For every $m > 0$ and $p > 0$, the flat menu (p, p) has the unique numerical cutoff $a = p$. Its completion probability is*

$$M_{F, G, \gamma}^{\text{flat}}(m, p) = [1 - G(p)](1 - e^{-mF(p)}) + e^{-mF(p)}[1 - \tilde{G}(p/\beta)](1 - e^{-\gamma mF(p)}). \quad (12)$$

Consequently

$$M_{F, G, \gamma}^{\text{flat},*}(m) = \max_{p \in [0, 1]} M_{F, G, \gamma}^{\text{flat}}(m, p) \quad (13)$$

is attained.

For $a < p$, the flat cutoff margin factors as

$$(p - a) [\phi(mF(a)) - \delta \alpha e^{-mF(a)} \rho_G(p) \phi(m\{\alpha[F(p) - F(a)] + \gamma F(p)\})] > 0.$$

Thus no lower cutoff is sustainable. At $a = p$, surviving rejectors have cost above p and cannot accept the same payment later; only fresh entrants can complete after failure. This gives (12), and continuity on the compact price interval gives attainment.

With no fresh entry, the objective simplifies to

$$M_{F, G, 0}^{\text{flat}}(m, p) = [1 - G(p)](1 - e^{-mF(p)}). \quad (\text{F3})$$

If F and G have positive densities and an optimizer is interior, its first-order condition is

$$\frac{g(p)}{1 - G(p)} = \frac{mf(p)}{e^{mF(p)} - 1}. \quad (\text{F4})$$

This hazard-versus-capacity equation is useful even when the optimal price has no closed form.

For later use under the uniform specialization $F(c) = c$, $G(v) = v$, write

$$M_{\gamma}^{\text{flat}}(m, p) = (1 - p)(1 - e^{-mp}) + e^{-mp}[1 - p/\beta]^+(1 - e^{-\gamma mp}),$$

$$M_{\gamma}^{\text{flat},*}(m) = \max_{p \in [0, 1]} M_{\gamma}^{\text{flat}}(m, p), \quad \mathcal{P}_{\gamma}^{\text{flat}}(m) = \arg \max_{p \in [0, 1]} M_{\gamma}^{\text{flat}}(m, p).$$

Let

$$p^0(m) = \frac{1 + m - W_0(e^{1+m})}{m}, \quad (\text{F1})$$

the unique maximizer of the first-window objective $(1 - p)(1 - e^{-mp})$.

LEMMA 2 (**Geometry of uniform optimized flat pricing**). *For $m > 0$, $\beta \in (0, 1)$, and $\gamma > 0$,*

$$\mathcal{P}_\gamma^{\text{flat}}(m) \subset (0, \beta) \cup \{p^0(m)\}, \quad 0 < p^0(m) < \frac{1}{2}. \quad (\text{F2})$$

The point $p^0(m)$ can be optimal only when $\beta < p^0(m)$. Hence, if $\beta \geq p^0(m)$ —in particular, if $\beta \geq 1/2$ —every optimized flat price is strictly below β . The optimizer correspondence need not be single valued.

The lemma prevents a hidden uniqueness assumption in the mixed-entry result. When $\beta < p^0(m)$, a low-price optimizer and $p^0(m)$ can tie for some $\gamma > 0$; the theorem below therefore requires one active optimizer, not a unique optimizer.

6. Rescue Pricing Beyond Uniform Distributions

The general equilibrium foundation permits a selection-robust local result. It preserves the central economic comparison without assuming a uniform cost or value distribution and without requiring nearby cutoff uniqueness.

THEOREM 3 (**Local rescue improvement**). *Suppose $\gamma = 0$, $0 < p < \beta$, and $\alpha > 0$. Let F be differentiable at p with $f(p) > 0$, and let G be continuous with full support. Define*

$$R = e^{-mF(p)}, \quad \sigma = \phi(mF(p)), \quad \rho_G(p) = \frac{1 - G(p/\beta)}{1 - G(p)}.$$

For the menu $(p, p + \varepsilon)$, every equilibrium cutoff satisfies, uniformly across equilibrium branches,

$$a_\varepsilon = p - \kappa(p)\varepsilon + o(\varepsilon), \quad \kappa(p) = \frac{\delta\alpha R\rho_G(p)}{\sigma - \delta\alpha R\rho_G(p)}. \quad (\text{G1})$$

Moreover,

$$\begin{aligned} & \min_{a_\varepsilon \in \mathcal{E}_{F,G}(p, p+\varepsilon)} M_{F,G}^{\alpha,0}(p, p+\varepsilon; a_\varepsilon) \\ &= M_{F,G,0}^{\text{flat}}(m, p) + L_{F,G}(p)\varepsilon + o(\varepsilon), \\ & L_{F,G}(p) = [1 - G(p)]mf(p)R \frac{\alpha\rho_G(p)[\sigma - \delta R]}{\sigma - \delta\alpha R\rho_G(p)} > 0. \end{aligned} \quad (\text{G2})$$

Consequently, if an optimal flat price p_F^ is strictly below β , then $(p_F^*, p_F^* + \varepsilon)$ strictly improves on the optimized flat policy for every sufficiently small $\varepsilon > 0$, even under worst-cutoff selection.*

The denominator in (G1) is positive because $\rho_G(p) < 1$ and $\phi(mF(p)) > e^{-mF(p)}$. The same inequality gives the strict numerator wedge $\sigma - \delta R$. Thus the local gain combines assignment competition, which is present at $\delta = 1$, with incumbent impatience, which strengthens the gain when $\delta < 1$. Fresh entry is excluded here because an entrant pool already creates terminal completion under the flat policy; Section 9 rederives the appropriate local condition for that case.

Convexity of the cost CDF turns the local statement into a global comparison for any active menu. Convex F means that successively higher cost intervals contain weakly more probability mass per unit of payment.

THEOREM 4 (Global same-payment dominance under convex supply). *Suppose $\gamma = 0$ and F is atomless, strictly increasing, and convex. Fix a strict menu $p_1 < p_2$ and an equilibrium cutoff $a < p_1$ for which a positive mass of riders activates rescue. Then*

$$M_{F,G}^{\alpha,0}(p_1, p_2; a) > [1 - G(p_1)]\{1 - e^{-mF(p_1)}\}. \quad (\text{G3})$$

Hence every active strict menu beats the flat policy with the same first payment, branch by branch; cutoff uniqueness is unnecessary.

To see the role of convexity, let

$$s_j(a) = \frac{F(p_j) - F(a)}{p_j - a}, \quad j \in \{1, 2\}.$$

Convexity gives $s_2(a) \geq s_1(a)$. Cutoff indifference then converts the terminal-coverage term into at least the supply mass lost below p_1 , while $a < p_1$ makes the comparison strict. Appendix EC.2 gives the complete inequality.

The condition is substantive. With

$$F(c) = \sqrt{c}, \quad G(v) = v, \quad (m, \alpha, \beta, \delta, \gamma) = (0.5, 0.75, 0.3, 1, 0),$$

the menu $(p_1, p_2) = (0.06, 0.17)$ has an equilibrium cutoff $a = 0.02031336$. Dynamic completion is 0.10348314, whereas the flat payment $p_1 = 0.06$ completes 0.10835523; a large rescue loses 0.4872 percentage points. This does not contradict Theorem 3, which concerns sufficiently small rescue increments.

General distributions also permit equilibrium multiplicity. Appendix EC.2 constructs continuous full-support F and G for which one menu has three transversal cutoffs, with completion probabilities 0.6926, 0.7398, and 0.7587. The 6.61 percentage-point selection spread explains why (10) uses a conservative minimum and why the unrestricted design problem retains a supremum. Neither multiplicity nor concavity affects the exact uniform no-entry results below.

7. Global menu design without fresh entry

We now impose the exact-core specialization

$$F(c) = c, \quad G(v) = v, \quad \gamma \geq 0, \quad m > 0, \quad \alpha \in (0, 1], \quad \beta \in (0, 1), \quad \delta \in (0, 1]. \quad (\text{U})$$

For compactness, write $\eta(a) = \eta^2(a)$, $\rho(p) = [1 - p/\beta]^+ / (1 - p)$, and use

$$\begin{aligned} \mathcal{E}_m^{\alpha,\delta,\gamma}(p_1, p_2) &:= \mathcal{E}_{F,G}(p_1, p_2), \\ M_m^{\alpha,\gamma}(p_1, p_2; a) &:= M_{F,G}^{\alpha,\gamma}(p_1, p_2; a), \\ \underline{M}_m^{\alpha,\delta,\gamma}(p_1, p_2) &:= \underline{M}_{F,G}^{\alpha,\delta,\gamma}(p_1, p_2), \\ D_{\alpha,\delta,\gamma}^*(m) &:= M_R^*(m; F, G, \alpha, \beta, \delta, \gamma) \quad \text{under (U)}. \end{aligned} \quad (\text{U1})$$

The notation in (U1) applies to the uniform model for arbitrary γ . For the global analysis in this section and Section 8, we now impose $\gamma = 0$. Uniformity together with no fresh entry produces the exact envelope and global one-dimensional reduction; it is not the source of cutoff existence. Let $k = \alpha m$. Menus with $p_1 \geq \beta$ or $p_2 \geq \beta$ cannot induce positive-mass rescue and are outcome-equivalent to the flat first payment. The nontrivial region is therefore $0 \leq p_1 < p_2 < \beta$.

For $a \in [0, p_1]$, put

$$C(x) = 1 - e^{-kx}, \quad h(a) = \begin{cases} (e^{ma} - 1)/a, & a > 0, \\ m, & a = 0, \end{cases} \quad (14)$$

and

$$A_z(a) = (\beta - z)C(z - a), \quad z \in \{p_1, p_2\}, \quad B_1 = \beta(1 - p_1). \quad (15)$$

The repeat-rescue switch lies below one exactly when $A_{p_2}(a) > A_{p_1}(a)$. Substitution of the switch type into continuation coverage gives the exact cancellation

$$K_{12}(a) = \frac{\max\{A_{p_1}(a), A_{p_2}(a)\}}{B_1}. \quad (16)$$

Multiplying the cutoff driver's payoff margin by the positive factor me^{ma} gives the sign-equivalent scalar

$$\mathcal{D}_{12}^\delta(a) = (p_1 - a)h(a) - \delta K_{12}(a). \quad (17)$$

The repeat branch cannot contain a cutoff below p_1 :

$$(p_1 - a)h(a) > \frac{A_{p_1}(a)}{B_1} \geq \delta \frac{A_{p_1}(a)}{B_1}, \quad 0 \leq a < p_1. \quad (18)$$

For $a > 0$, this is $\phi(ma) > e^{-ma}$ combined with $\alpha\rho(p_1)\phi(k(p_1 - a)) \leq 1$; at $a = 0$, strictness follows from $\rho(p_1) < 1$. Hence every zero of (17) under a strict active menu lies on the rescue branch, not at the rider kink.

THEOREM 5 (Global cutoff uniqueness). *Fix $0 \leq p_1 < p_2 < \beta$. If $\mathcal{D}_{12}^\delta(0) \leq 0$, the unique equilibrium cutoff is the lower-boundary cutoff $a = 0$. If $\mathcal{D}_{12}^\delta(0) > 0$, there is a unique cutoff $a \in (0, p_1)$. Equality at zero never coexists with a positive root. Together with the flat, repeat-only, $p_1 = 0$, and $\alpha = 0$ boundary cases, every no-entry menu has a unique symmetric equilibrium cutoff under the stated cutoff-type convention.*

The one-crossing argument is global. At an active-rescue zero write $u = p_1 - a$, $z = p_2 - a$, $b = \beta - p_2$, and $B = \beta(1 - p_1)$. The root equation is

$$uh(a) = \delta \frac{b}{B} C(z).$$

Since $B - b = p_2 - \beta p_1 > 0$ and $h(a) \geq m$, $u < C(z)/m$. For $R(a) = uh(a)/C(z)$,

$$\frac{R'(a)}{R(a)} = -\frac{1}{u} + \frac{h'(a)}{h(a)} + \frac{k}{e^{kz} - 1} < 0, \quad (19)$$

because

$$\frac{1}{u} - \frac{k}{e^{kz} - 1} > m \frac{1 - \alpha e^{-kz}}{1 - e^{-kz}} \geq m > \frac{h'(a)}{h(a)}.$$

Thus every zero crosses downward. Since $\mathcal{D}_{12}^\delta(p_1) < 0$, two zeros would require an intervening upward crossing. At $p_1 = 0$, the numeric cutoff is zero and the cost-zero type uses the lower-boundary reject action.

7.1. Completion identity and comparison with the same flat payment

At any positive cutoff, indifference gives

$$e^{-ma} K_{12}(a) = \frac{m\phi(ma)(p_1 - a)}{\delta}.$$

Using $1 - e^{-ma} = m\phi(ma)$ in (9) yields

$$M_m^{\alpha, \delta, 0}(p_1, p_2; a) = (1 - p_1)m\phi(ma) \left[a + \frac{p_1 - a}{\delta} \right], \quad a > 0. \quad (20)$$

At a reject-all equilibrium,

$$M_m^{\alpha, \delta, 0}(p_1, p_2; 0) = \left(1 - \frac{p_2}{\beta} \right) (1 - e^{-kp_2}) \geq \frac{(1 - p_1)mp_1}{\delta}, \quad (21)$$

with equality exactly at the lower-boundary indifference condition. Repeat may be used by positive mass in (21); its contribution cancels algebraically.

Both $\phi(ma)$ and $a + (p_1 - a)/\delta$ weakly decrease in $a > 0$, and the first decreases strictly, so (20) decreases with the cutoff. Theorem 5 then gives a particularly strong comparison. For every $0 < p_1 < p_2 < \beta$, the strict-menu cutoff is below p_1 , and therefore

$$\underline{M}_m^{\alpha, \delta, 0}(p_1, p_2) > F_m(p_1) = (1 - p_1)(1 - e^{-mp_1}). \quad (22)$$

This is a global same- p_1 result, not merely a derivative along one selected branch.

For completeness, a one-sided local expansion at (p_1, p_1) is

$$a_\varepsilon = p_1 - \kappa\varepsilon + o(\varepsilon), \quad \kappa = \frac{\delta e^{-mp_1} \alpha \rho(p_1)}{\phi(mp_1) - \delta e^{-mp_1} \alpha \rho(p_1)}, \quad (23)$$

and

$$\left. \frac{dM(p_1, p_1 + \varepsilon)}{d\varepsilon} \right|_{0+} = \frac{m(1 - p_1)e^{-mp_1} \alpha \rho(p_1) [\phi(mp_1) - \delta e^{-mp_1}]}{\phi(mp_1) - \delta e^{-mp_1} \alpha \rho(p_1)} > 0. \quad (24)$$

The wedge decomposes as $\phi(mp_1) - \delta e^{-mp_1} = [\phi(mp_1) - e^{-mp_1}] + (1 - \delta)e^{-mp_1}$: assignment competition is the first term and incumbent impatience the second.

7.2. The fixed-first-payment problem

Following the design sequence, fix p_1 before optimizing p_2 . For a target cutoff $a < \beta$, maximize

$$A(a, p_2) = (\beta - p_2)[1 - e^{-k(p_2 - a)}], \quad p_2 \in [a, \beta]. \quad (25)$$

It is strictly concave. Its unique maximizer $Q(a)$ satisfies

$$e^{k(Q(a) - a)} - 1 = k(\beta - Q(a)), \quad (26)$$

or

$$Q(a) = \beta - \frac{W_0(e^{1+k(\beta-a)}) - 1}{k}. \quad (27)$$

Moreover

$$a < Q(a) < \frac{a + \beta}{2}.$$

Let

$$S(a) = A(a, Q(a)), \quad S(a) = \frac{k[\beta - Q(a)]^2}{1 + k[\beta - Q(a)]}. \quad (28)$$

Define $Q_0 = Q(0)$, $S_0 = S(0)$, and

$$p_z = \frac{1 - \sqrt{1 - 4\delta S_0/(\beta m)}}{2}. \quad (29)$$

The strict flat inequality at $p_1 = Q_0$, $a = 0$ gives $0 < p_z < Q_0 < 1/2$.

THEOREM 6 (Optimal rescue conditional on the first payment). *Let*

$$H_m(p_1) = \max_{p_2 \in [p_1, 1]} \underline{M}_m^{\alpha, \delta, 0}(p_1, p_2).$$

Then:

- (i) *If $0 \leq p_1 \leq p_z$, the unique optimal rescue is $p_2 = Q_0$, the cutoff is the lower-boundary reject-all cutoff $a = 0$, and*

$$H_m(p_1) = S_0/\beta.$$

- (ii) *If $p_z < p_1 < \beta$, there is a unique $a^*(p_1) \in (0, p_1)$ satisfying*

$$(p_1 - a)h(a) = \frac{\delta S(a)}{\beta(1 - p_1)}. \quad (30)$$

The unique optimal rescue is $p_2 = Q(a^(p_1))$, and*

$$H_m(p_1) = (1 - p_1)m\phi(ma^*(p_1)) \left[a^*(p_1) + \frac{p_1 - a^*(p_1)}{\delta} \right].$$

- (iii) *If $\beta \leq p_1 \leq 1$, every $p_2 \in [p_1, 1]$ is outcome-equivalent and*

$$H_m(p_1) = (1 - p_1)(1 - e^{-mp_1}).$$

Thus the conditional supremum is a maximum in every regime.

To see implementability, a positive target cutoff solves

$$A(a, p_2) = \frac{\beta(1-p_1)(p_1-a)h(a)}{\delta}. \quad (31)$$

The right side strictly exceeds $A(a, p_1)$ by (18), so (31) cannot generate a repeat-only or kink pseudo-solution. If the envelope is below, equal to, or above the right side, there are respectively zero, one tangent, or two strict implementing prices. Theorem 5 makes each implemented cutoff the unique equilibrium. Completion falls with the cutoff, so the smallest implementable cutoff, attained at tangency, solves the fixed- p_1 problem.

7.3. One-dimensional global design and attainment

Define

$$R_\delta(a) = \frac{\delta S(a)}{\beta h(a)}.$$

The tangent equation is

$$(1-p_1)(p_1-a) = R_\delta(a). \quad (32)$$

Because $Q(a) < (1+a)/2$ and the left side is increasing in p_1 below that vertex, the unique feasible root is the lower quadratic branch

$$P_\delta(a) = \frac{1+a - \sqrt{(1-a)^2 - 4R_\delta(a)}}{2}. \quad (33)$$

The upper branch is not a fixed- p_1 rescue optimum. The map P_δ is continuous and strictly increasing from $[0, \beta]$ onto $[p_z, \beta]$. Endpoint injectivity follows from the same envelope one-crossing argument: $P_\delta(a) = P_\delta(0) = p_z$ for $a > 0$ would give the fixed- $p_1 = p_z$ envelope two zeros.

Extend $P_\delta(\beta) = Q(\beta) = \beta$ and set

$$J_\delta(a) = \begin{cases} (1-P_\delta(a))m\phi(ma) \left[a + \frac{P_\delta(a)-a}{\delta} \right], & a > 0, \\ mp_z(1-p_z)/\delta = S_0/\beta, & a = 0. \end{cases} \quad (34)$$

Then $J_\delta(\beta) = (1-\beta)(1-e^{-m\beta})$, and the complete outer problem is

$$D_{\alpha, \delta, 0}^*(m) = \max \left\{ \max_{a \in [0, \beta]} J_\delta(a), \max_{p \in [\beta, 1]} (1-p)(1-e^{-mp}) \right\}. \quad (35)$$

THEOREM 7 (Policy attainment and optimized-flat dominance). *For every no-entry environment with positive m and positive α , the announced policy value in (11) is attained. If $\beta \geq 1/2$, every optimal menu is strict:*

$$(p_1^*, p_2^*) = (P_\delta(a^*), Q(a^*)), \quad a^* \in \arg \max_{a \in [0, \beta]} J_\delta(a) \subset (0, \beta), \quad (36)$$

and

$$D_{\alpha, \delta, 0}^*(m) > F_0^*(m). \quad (37)$$

No uniqueness of a^* is asserted.

Equation (35) is a maximum of continuous functions on compact intervals, so it proves attainment directly. In the no-entry flat problem,

$$F_m(p) = (1 - p)(1 - e^{-mp})$$

is strictly concave and its unique optimizer is

$$p_F(m) = \frac{1 + m - W_0(e^{1+m})}{m} < \frac{1}{2}. \quad (38)$$

For $\beta \geq 1/2$, $p_F < \beta$, and (22) applied at p_F proves (37), including the equality case $\beta = 1/2$. The boundary portions of (35) do not beat F_0^* , whereas (37) does, so every global optimizer is interior and strict.

The driver discount also yields a clean comparative static. Write $R_1(a) = S(a)/[\beta h(a)]$, so that $(1 - P_\delta)(P_\delta - a) = \delta R_1(a)$. For $a \in (0, \beta)$,

$$\frac{\partial P_\delta(a)}{\partial \delta} = \frac{R_1(a)}{1 + a - 2P_\delta(a)} > 0, \quad \frac{\partial J_\delta(a)}{\partial \delta} = -(1 - e^{-ma}) \frac{\partial P_\delta(a)}{\partial \delta} < 0. \quad (38a)$$

Thus $D_{\alpha, \delta, 0}^*(m)$ is weakly decreasing in δ : more patient incumbents require a higher first payment to implement the same cutoff and continuation coverage. This is a completion comparative static, not a welfare ranking.

7.4. The full patience region

Fix (m, α, δ) and make patience explicit in $D(m; \beta)$. For a fixed strict menu $p_1 < p_2$, its value equals $F_m(p_1)$ for $\beta \leq p_2$ and is strictly increasing for $\beta > p_2$. Indeed, increasing β raises the active-rescue coefficient in (17), shifts the unique cutoff strictly left (or to zero), and then raises completion by (20)–(21).

The scalar representation is jointly continuous in patience. To use a fixed choice set, write $a = \beta x$. The solution of

$$e^{ky} - 1 = k\{\beta(1 - x) - y\}$$

is continuous in (β, x) . At $x = 0$, $h(0) = m$ and $P_{\beta, \delta}(0) = p_z(\beta)$ has a strictly positive discriminant; at $x = 1$, $P_{\beta, \delta}(\beta) = Q_\beta(\beta) = \beta$ and $J_{\beta, \delta}(\beta) = (1 - \beta)(1 - e^{-m\beta})$. These removable extensions and the elementary maximum theorem make $D(m; \beta)$ continuous.

Define

$$\beta_c(m, \alpha, \delta) = \inf\{\beta \in (0, 1) : D_{\alpha, \delta, 0}^*(m; \beta) > F_0^*(m)\}. \quad (39)$$

THEOREM 8 (Patience threshold). *For $m > 0$, $0 < \alpha \leq 1$, and $0 < \delta \leq 1$, the strict-gain region is the single upper interval*

$$D_{\alpha, \delta, 0}^*(m; \beta) > F_0^*(m) \iff \beta > \beta_c(m, \alpha, \delta). \quad (40)$$

Moreover

$$0 < -\frac{1}{m} \log(1 - F_0^*(m)) \leq \beta_c(m, \alpha, \delta) \leq p_F(m) < \frac{1}{2}. \quad (41)$$

For fixed (m, α, δ) , $D(m; \beta) = F_0^*(m)$ through the threshold and is strictly increasing above it. The exact scalar test is

$$\beta > \beta_c(m, \alpha, \delta) \iff \max_{a \in [0, \beta]} J_{\beta, \delta}(a) > F_0^*(m). \quad (42)$$

When $\alpha = 0$, the gain region is empty.

Pointwise monotonicity in (38a) also nests the thresholds:

$$0 < \delta_1 < \delta_2 \leq 1 \implies \beta_c(m, \alpha, \delta_1) \leq \beta_c(m, \alpha, \delta_2). \quad (42a)$$

Strict inequality is not asserted globally.

The lower bound follows because an active menu has $p_2 < \beta$ and completion requires an initial incumbent with cost at most p_2 , giving $M < 1 - e^{-m\beta}$. The upper bound follows from a strict menu at p_F when $\beta > p_F$. If gain holds at one patience level, attainment supplies an optimal strict menu; reusing it at any larger β and applying the fixed menu monotonicity gives strict gain and strict value increase. Continuity makes the gain interval open.

8. Market thickness: topology and endpoint rates

Fix $0 < \alpha \leq 1$, $0 < \beta < 1$, $0 < \delta \leq 1$, and $\gamma = 0$. Holding these primitives and the same thickness m fixed, define

$$M_R^*(m) := D_{\alpha, \delta, 0}^*(m), \quad M_F^*(m) := F_0^*(m), \quad V(m) := M_R^*(m) - M_F^*(m). \quad (43)$$

Thus $V(m)$ is the completion-rate gain from allowing an optimally designed rescue menu instead of an optimally chosen flat payment at the same market thickness. It is not the marginal value of adding drivers; $100V(m)$ is the gain in percentage points. The outer representation (35), rather than a local-menu derivative, is the appropriate object because the optimal menu moves with thickness.

For the thin limit, define on $a \in [0, \beta]$

$$P_0^\delta(a) = \frac{1 + a - \sqrt{(1-a)^2 - \delta\alpha(\beta-a)^2/\beta}}{2}, \quad H_0^\delta(a) = a[1 - P_0^\delta(a)] + \frac{\alpha(\beta-a)^2}{4\beta}, \quad (44)$$

and $\mathcal{H}_{\alpha, \beta, \delta} = \max_{a \in [0, \beta]} H_0^\delta(a)$.

THEOREM 9 (Thickness topology and exact endpoint regimes). Fix $0 < \alpha \leq 1$, $0 < \beta < 1$, and $0 < \delta \leq 1$.

- (i) $M_R^*(m)$, $M_F^*(m)$, and $V(m)$ are continuous on $(0, \infty)$. The scalar cutoff and flat-price argmax correspondences are nonempty, compact valued, and upper hemicontinuous.

(ii) If $\beta > 1/2$ and $\delta < 1$, then

$$V(m) \sim \left(\mathcal{H}_{\alpha,\beta,\delta} - \frac{1}{4} \right) m, \quad \mathcal{H}_{\alpha,\beta,\delta} > \frac{1}{4}. \quad (45)$$

At the no-discount boundary $\delta = 1$, let $a_0 \in (0, 1/2)$ be the unique solution of

$$\alpha(\beta - a_0)^2 = \beta(1 - 2a_0).$$

Then

$$V(m) \sim \frac{1 - 2a_0}{16} m^2. \quad (46)$$

(iii) If $\beta = 1/2$ and $\delta < 1$, then

$$V(m) \sim \frac{\alpha(1 - \delta)}{512\{1 - \alpha(1 - \delta)/2\}} m^3. \quad (47)$$

At $\delta = 1$, the leading cancellation persists one order longer:

$$V(m) \sim \frac{\alpha}{2048} m^4. \quad (48)$$

(iv) If $\beta < 1/2$, there exists $m_0(\alpha, \beta, \delta) > 0$ such that

$$M_R^*(m) = M_F^*(m), \quad V(m) = 0, \quad 0 < m < m_0(\alpha, \beta, \delta). \quad (49)$$

(v) As $m \rightarrow \infty$,

$$1 - M_R^*(m) \sim \frac{\log \log m}{m}, \quad 1 - M_F^*(m) \sim \frac{\log m}{m}, \quad V(m) \sim \frac{\log m}{m}. \quad (50)$$

For each fixed positive (α, β) , the estimates establishing (50) are uniform over the full policy/cutoff domain and uniformly over $\delta \in (0, 1]$. They are not uniform over primitive sequences with $\alpha \downarrow 0$ or $\beta \downarrow 0$.

Equations (45)–(48) show a singular boundary: the limits $m \downarrow 0$ and $\delta \uparrow 1$ do not commute at the first nonzero order. Even slight incumbent impatience creates a first-order screening gain when $\beta > 1/2$ and a cubic gain at $\beta = 1/2$; exact patience removes those terms.

Here is the proof architecture. Let $k = \alpha m$ and let $z_m(a)$ solve

$$z_m(a) + \log[1 + z_m(a)] = k(\beta - a), \quad y_m(a) = z_m(a)/k.$$

Then

$$Q_m(a) = \beta - y_m(a), \quad T_m(a) = \frac{k y_m(a)^2}{\beta[1 + k y_m(a)]}, \quad B_m^\delta(a) = \delta \frac{a}{e^{ma} - 1} T_m(a), \quad (51)$$

with $a/(e^{ma} - 1) = 1/m$ at zero, and

$$P_m^\delta(a) = \frac{1 + a - \sqrt{(1 - a)^2 - 4B_m^\delta(a)}}{2}. \quad (52)$$

The scalar objective admits the useful exact form

$$J_m^\delta(a) = [1 - P_m^\delta(a)](1 - e^{-ma}) + T_m(a)e^{-ma}. \quad (53)$$

The strict envelope bound keeps the discriminant positive. Joint continuity and Berge's maximum theorem prove part (i).

Uniformly on $[0, \beta]$, $P_m^\delta \rightarrow P_0^\delta$ and $J_m^\delta/m \rightarrow H_0^\delta$. If $\beta > 1/2$ and $\delta < 1$, the unique $a \in (0, 1/2)$ satisfying $P_0^\delta(a) = 1/2$ gives

$$H_0^\delta(a) = \frac{1}{2} \left[a + \frac{1/2 - a}{\delta} \right] > \frac{1}{4},$$

which proves the strict coefficient in (45); uniform convergence gives its exact value as the maximum in (44). For $\delta = 1$, the first-order term ties the flat benchmark, and the uniform second-order argmax expansion gives (46).

At $\beta = 1/2$, every thin optimizer localizes as $a = 1/2 - cm$. Put $A_\delta = 1 - \alpha(1 - \delta)/2$. Uniformly for bounded c ,

$$J_m^\delta(1/2 - cm) = \frac{m}{4} - \frac{m^2}{16} + m^3 \left\{ \frac{1}{96} + \frac{c}{8} - A_\delta c^2 \right\} + O(m^4). \quad (54)$$

The maximizing rescaled cutoff is $c^* = 1/(16A_\delta)$, while the flat m^3 coefficient is $11/768$. Their difference is exactly the coefficient in (47). When $\delta = 1$ it vanishes; the fourth-order expansion yields (48). In particular, uniformly for bounded c ,

$$\begin{aligned} J_m^1(1/2 - cm) &= \frac{m}{4} - \frac{m^2}{16} + m^3 f_2(c) + m^4 f_3(c) + o(m^4), \\ f_2(c) &= \frac{11}{768} - \left(c - \frac{1}{16} \right)^2, \\ f_3(c) &= -\frac{1}{768} - \frac{c}{24} + \frac{c^2}{4} + 2\alpha c^3, \end{aligned} \quad (54a)$$

whereas

$$F_0^*(m) = \frac{m}{4} - \frac{m^2}{16} + \frac{11}{768}m^3 - \frac{3}{1024}m^4 + o(m^4). \quad (54b)$$

The strict quadratic loss in f_2 and local Lipschitz control of f_3 evaluate the perturbed maximum at $c = 1/16$ to fourth order, and $f_3(1/16) = -3/1024 + \alpha/2048$. Subtracting (54b) proves (48). If $\beta < 1/2$, the bound

$$H_0^\delta(a) \leq a(1 - a) + \frac{(\beta - a)^2}{4\beta} \leq \beta(1 - \beta) < \frac{1}{4}$$

is uniform on $[0, \beta]$. Since the flat optimizer converges to $1/2$, the flat policy wins exactly on a nonempty initial interval, proving (49).

For the thick limit put $x = ma$, $p_1 = P_m^\delta(a)$, and $T = T_m(a)$. Equation (53) gives the same exact nonnegative loss decomposition for every fixed $\delta > 0$:

$$1 - J_m^\delta(a) = p_1(1 - e^{-x}) + (1 - T)e^{-x}. \quad (55)$$

Table 2 Selected-thickness theoretical solutions.

m	p_1^*	p_2^*	a^*	M_R^*	M_F^*	$100V(m)$
1	0.3919	0.5464	0.3203	0.2132	0.1993	1.3883
5	0.2369	0.4131	0.1978	0.5978	0.5438	5.4003
10	0.1559	0.3103	0.1329	0.7552	0.6936	6.1552
20	0.0925	0.2081	0.0804	0.8624	0.8080	5.4437

Parameters are $\alpha = 1$, $\beta = \delta = 0.8$, and $\gamma = 0$. a^* is the equilibrium first-period cutoff and $100V(m)$ is the completion gain in percentage points.

The feasible choice $x = \log \log m$ and the policy-uniform lower bound obtained by splitting at $Q_m(a) = \beta/2$ give the first rate in (50). The exact flat first-order condition gives the second, and subtraction gives the third.

COROLLARY 1 (A best finite thickness, without a shape theorem). *For every fixed positive (α, β, δ) in the maintained ranges, there exists $m^* \in (0, \infty)$ such that*

$$V(m^*) = \max_{m > 0} V(m) > 0. \quad (56)$$

Strict single-peakedness is false when $\beta < 1/2$ because of (49). For $\beta \geq 1/2$, strict single-peakedness remains open for every fixed $\delta \in (0, 1]$.

Continuity, endpoint vanishing, and eventual positivity in (50) localize a global maximizer to a compact subinterval. No uniqueness of m^* is claimed.

8.1. Theoretical numerical illustration

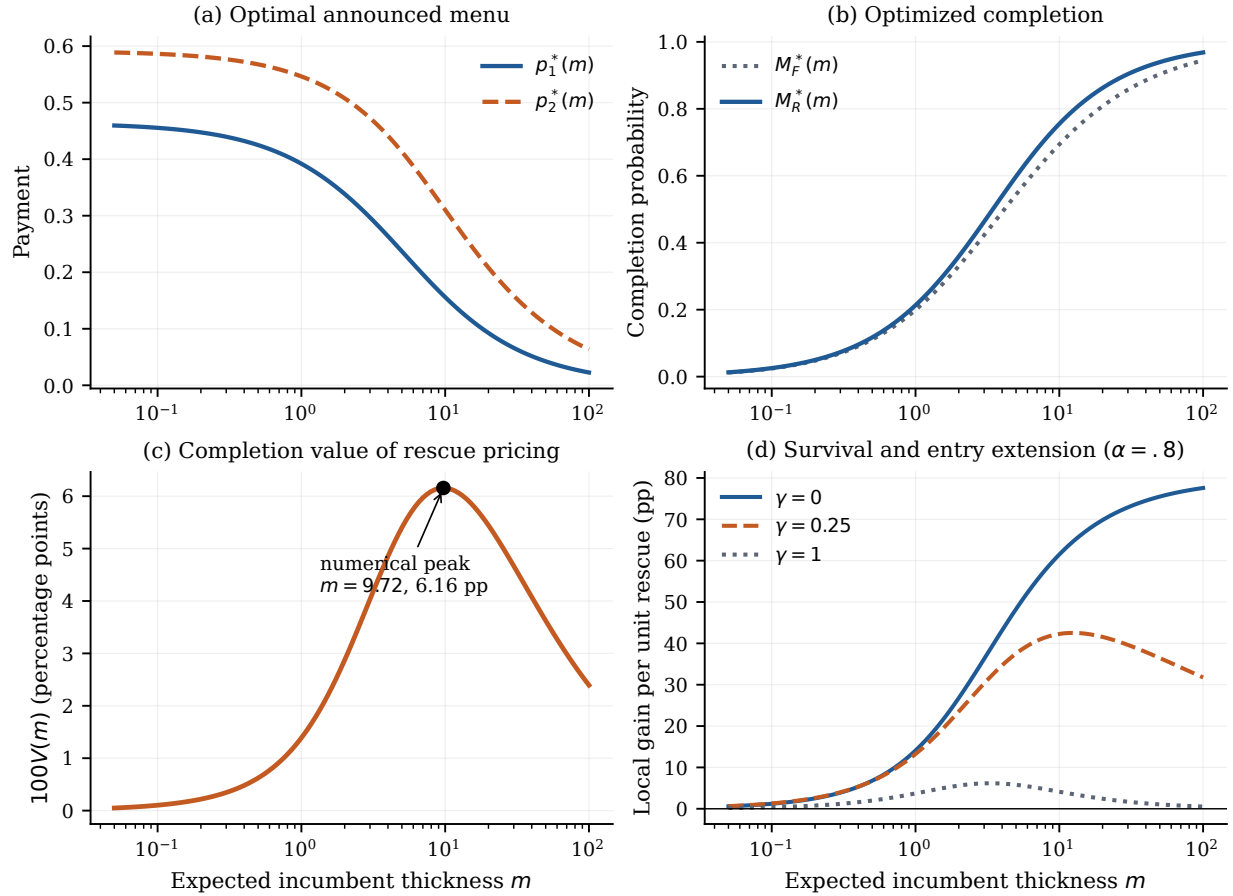
Figure 2 and Table 2 illustrate the uniform model at $\alpha = 1$ and $\beta = \delta = 0.8$. These values are chosen to display the mechanism. Each thickness point re-solves the equilibrium cutoff and then optimizes p_2 conditional on p_1 before optimizing p_1 . Figure 3 repeats that full optimization while varying rider patience, incumbent patience, and survival. It is therefore a policy-class comparison, rather than the response to holding one menu fixed.

Figure 4 visualizes two boundaries that the baseline profile alone cannot show. Panel (a) uses the exact concave-CDF example from Section 6. A small rescue is beneficial, as the local theorem predicts, but the comparison reverses once the rescue increase is large. Panel (b) plots the mixed-supply activity condition from Section 9; fresh entry can make repeating the old payment so effective that no positive rider mass activates a marginal rescue.

9. Extension: Surviving Incumbents and Fresh Driver Entry

We retain the uniform type distributions from the exact core and reintroduce fresh entry. The purpose is to isolate how screening history changes the composition of continuation supply; the results are local rather than a global solution of the mixed-supply menu problem.

Figure 2 Theoretical rescue profiles and the survival–entry extension. Panels (a)–(c) use $\alpha = 1$, $\beta = \delta = 0.8$, and $\gamma = 0$. The marked point in panel (c) is a numerical maximizer, not a uniqueness theorem. Panel (d) fixes $\alpha = 0.8$, evaluates a marginal announced rescue at the no-entry flat benchmark, and varies fresh-entry intensity γ . The dashed and dotted profiles have $0 < \alpha < 1$ and $\gamma > 0$, so the terminal pool contains both screened surviving incumbents and unscreened new drivers; the solid $\gamma = 0$ profile is the nested no-entry boundary. Zero local gain can reflect an inactive rescue mass.



After universal rejection, two economically distinct sources of terminal supply coexist. A surviving incumbent appears with probability α , has already revealed a cost above the first-period cutoff, and discounts the payoff from waiting by δ . A fresh entrant arrives at intensity γm , has not been screened by the first response, and evaluates the terminal payment contemporaneously. The resulting eligible intensity at p_j is

$$m\{\alpha(p_j - a) + \gamma p_j\},$$

which records screening history rather than collapsing continuation supply to the single total $(\alpha + \gamma)m$. This composition effect is the focus of the extension. The central mixed-continuation case has $0 < \alpha < 1$ and $\gamma > 0$: some rejected incumbents disappear, some remain, and new drivers enter before the terminal response window. The cases $\gamma = 0$ or $\alpha \in \{0, 1\}$ are nested boundaries, not substitutes for the mixed case.

Figure 3 Optimized rescue gains across demand and supply primitives. Each curve re-solves the equilibrium cutoff and both payments at every thickness. Unless varied in a panel, parameters are $\alpha = 1$ and $\beta = \delta = 0.8$, with $\gamma = 0$. Higher rider patience and incumbent survival enlarge the attainable gain; greater incumbent patience reduces it because waiting becomes more attractive.

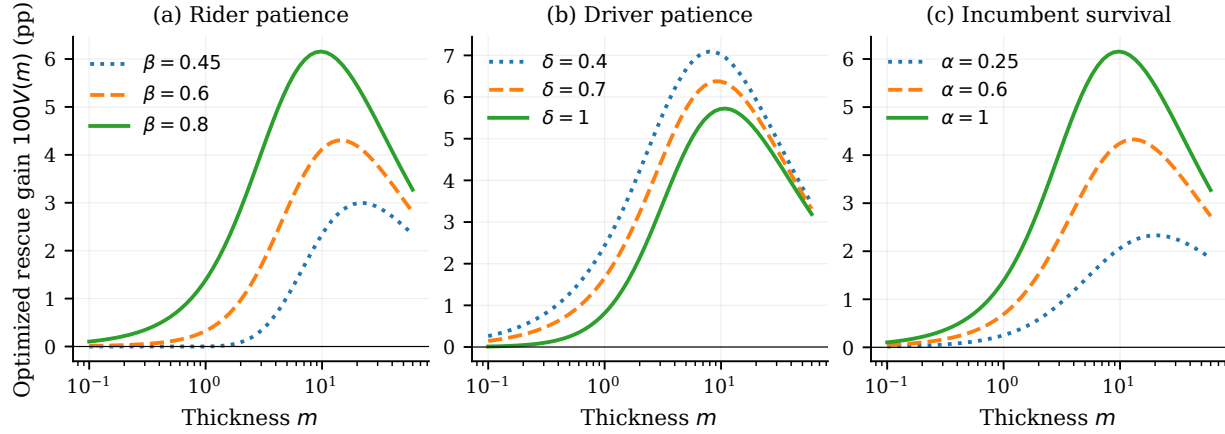
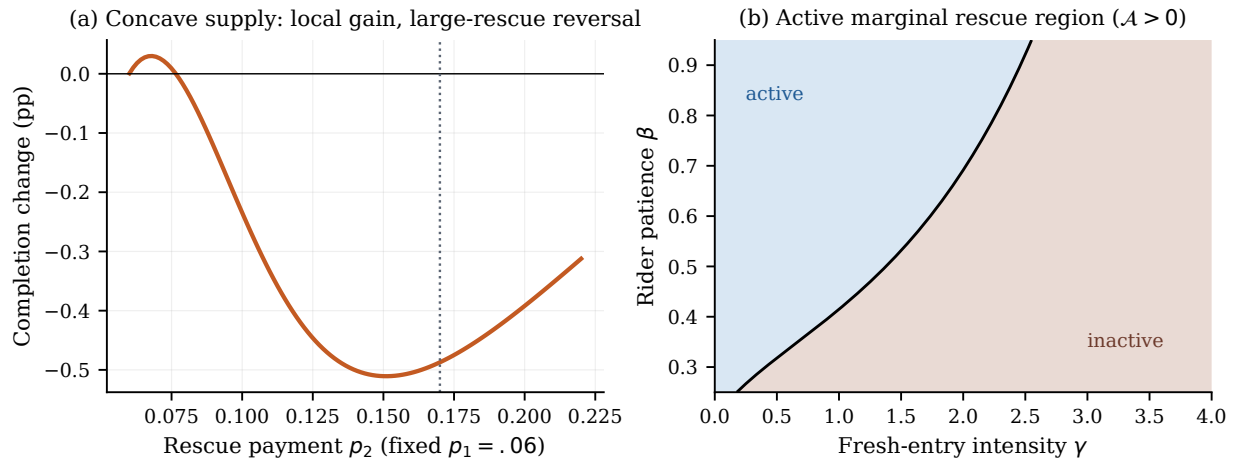


Figure 4 Where the rescue comparison changes sign or becomes inactive. Panel (a) fixes $F(c) = \sqrt{c}$, $G(v) = v$, $(m, \alpha, \beta, \delta, \gamma) = (.5, .75, .3, 1, 0)$, and $p_1 = .06$; the dotted line marks $p_2 = .17$. The plotted value is the conservative minimum across numerical cutoff roots. Panel (b) plots $\mathcal{A}_{\alpha, \gamma}(p)$ at $m = 5$, $p = .2$, and $\alpha = .6$; the blue region satisfies the strict activity condition.



The general equilibrium characterization permits a sharp local comparison with both sources present. Fix $0 < p < \beta$ and abbreviate

$$x = mp, \quad R = e^{-x}, \quad E = e^{-\gamma x}, \quad \sigma = \phi(x), \quad \ell = \phi(\gamma x), \quad \rho = \frac{1 - p/\beta}{1 - p}. \quad (\text{L1})$$

For $\alpha + \gamma > 0$, define the limiting repeat-rescue switch and rescue mass by

$$\bar{v} = \frac{1}{\beta} \left[p + \frac{1 - E}{mE(\alpha + \gamma)} \right], \quad \eta^0 = \frac{1 - \bar{v}}{1 - p}. \quad (\text{L2})$$

At $\gamma = 0$, the fraction in (L2) is zero.

THEOREM 10 (Fresh-entry local rescue). *Suppose $m > 0$, $0 < p < \beta < 1$, $0 < \delta \leq 1$, $0 \leq \alpha \leq 1$, $\gamma \geq 0$, $\alpha + \gamma > 0$, and $\bar{v} < 1$. For all sufficiently small $\varepsilon > 0$, the menu $(p, p + \varepsilon)$ has a unique anonymous symmetric equilibrium cutoff, up to behavior of the null cutoff type, with*

$$a_\varepsilon = p - \kappa\varepsilon + o(\varepsilon), \quad \kappa = \frac{\delta R \alpha \ell \eta^0}{\sigma - \delta R \alpha \rho \ell}. \quad (\text{L3})$$

Every candidate cutoff is uniformly localized by

$$0 \leq p - a_\varepsilon \leq \frac{\delta \alpha \rho}{1 - \delta \alpha \rho} \varepsilon. \quad (\text{L4})$$

The right derivative of completion is

$$\left. \frac{dM_m^{\alpha, \gamma}(p, p + \varepsilon; a_\varepsilon)}{d\varepsilon} \right|_{0+} = m(1 - p) R \eta^0 \frac{T_\delta}{\sigma - \delta R \alpha \rho \ell} > 0, \quad (\text{L5})$$

where

$$T_\delta = E(\alpha + \gamma)\sigma - \delta R \alpha \ell \{1 - \rho + \rho E(1 + \gamma)\} > 0. \quad (\text{L6})$$

The activity condition $\bar{v} < 1$ says that a positive mass of riders uses marginal rescue after failure. It is automatic at $\gamma = 0$ but substantive with entry. The knife edge $\bar{v} = 1$ needs a second-order rider-mass expansion and is not included. For $\gamma > 1$, activity cannot simply be deleted from the sign claim: for example

$$x = 1, \quad p = \frac{1}{2}, \quad m = 2, \quad \beta = \frac{51}{100}, \quad \alpha = 1, \quad \gamma = 10, \quad \delta = \frac{1}{2}$$

has $\bar{v} > 1$ and makes T_δ in (L6) negative. Thus the theorem's restriction is economically and mathematically real.

For $p < \beta$, define the activity index

$$\mathcal{A}_{\alpha, \gamma}(p) = m e^{-\gamma m p} (\alpha + \gamma)(\beta - p) - (1 - e^{-\gamma m p}). \quad (\text{L7})$$

Then $\mathcal{A}_{\alpha, \gamma}(p) > 0$ is equivalent to $\bar{v} < 1$, or equivalently

$$\alpha > \frac{e^{\gamma m p} - 1}{m(\beta - p)} - \gamma. \quad (\text{L8})$$

THEOREM 11 (Optimized-flat dominance with mixed continuation supply). *Fix positive m and γ , let $0 < \alpha < 1$, $0 < \beta < 1$, and $0 < \delta \leq 1$. Suppose that some $p^* \in \mathcal{P}_\gamma^{\text{flat}}(m)$ satisfies*

$$p^* < \beta, \quad \mathcal{A}_{\alpha, \gamma}(p^*) > 0. \quad (\text{L9})$$

For every sufficiently small $\varepsilon > 0$, the menu $(p^, p^* + \varepsilon)$ has a unique anonymous symmetric equilibrium cutoff, up to behavior of the null cutoff type, and*

$$\underline{M}_m^{\alpha, \delta, \gamma}(p^*, p^* + \varepsilon) = M_\gamma^{\text{flat}, *}(m) + \Lambda(p^*)\varepsilon + o(\varepsilon), \quad \Lambda(p^*) > 0. \quad (\text{L10})$$

Consequently,

$$D_{\alpha,\delta,\gamma}^*(m) > M_{\gamma}^{\text{flat},*}(m). \quad (\text{L11})$$

Neither uniqueness of the flat optimizer nor attainment of the general dynamic-policy supremum is required.

If instead $p < \beta$ and $\mathcal{A}_{\alpha,\gamma}(p) < 0$, then for every sufficiently small $\varepsilon > 0$ the same-price perturbation $(p, p + \varepsilon)$ has the unique cutoff $a = p$, no rider activates rescue, and its completion equals $M_{\gamma}^{\text{flat}}(m, p)$ exactly. No first-order claim is made at the knife edge $\mathcal{A}_{\alpha,\gamma}(p) = 0$.

The coefficient $\Lambda(p^*)$ is the expression in (L5), evaluated at $p = p^*$. Condition (L9) has a direct behavioral interpretation: after flat failure, even the highest rider type must strictly prefer access to the marginally broader rescue pool over repeating p^* . More fresh entrants do not make this condition automatic, because entry also raises repeat coverage.

Two different quantities can reasonably be called “the amount” of continuation supply. Conditional on universal rejection at the flat cutoff $a = p$, the expected number of second-window potential drivers is

$$m\tau_p, \quad \tau_p = \alpha(1 - p) + \gamma. \quad (\text{L12})$$

By contrast, holding the cutoff fixed, the number made newly eligible by a small payment increase has slope

$$\left. \frac{1}{m} \frac{\partial \lambda_2}{\partial p_2} \right|_{a=p} = \alpha + \gamma. \quad (\text{L13})$$

The next result fixes each measure in turn at the flat benchmark. These are local comparisons, not equal-baseline optimized-policy rankings. Along the equilibrium response $a_{\varepsilon} = p - \kappa\varepsilon + o(\varepsilon)$, raw headcount changes at rate $m\alpha\kappa$, and terminal eligible intensity changes at rate $m\{\alpha(1 + \kappa) + \gamma\}$. The composition response is precisely why neither baseline normalization removes strategic waiting.

THEOREM 12 (Supply composition and strategic waiting). *Fix $m > 0$, $0 < p < \beta < 1$, and $\delta \in (0, 1]$, and consider $(p, p + \varepsilon)$.*

(i) Fixed continuation headcount. Fix $h > 0$ and set

$$\gamma_h(\alpha) = h - (1 - p)\alpha, \quad A_h(\alpha) = \alpha + \gamma_h(\alpha) = h + p\alpha \quad (\text{L14})$$

for $0 \leq \alpha \leq \min\{1, h/(1 - p)\}$. This holds the conditional headcount in (L12) fixed at mh . Define

$$\ell_h(\alpha) = \phi(\gamma_h(\alpha)mp), \quad \eta_h(\alpha) = \frac{1}{1 - p} \left[1 - \frac{1}{\beta} \left\{ p + \frac{e^{\gamma_h(\alpha)mp} - 1}{mA_h(\alpha)} \right\} \right]. \quad (\text{L15})$$

On every interval on which $\eta_h(\alpha) > 0$, the unique local cutoff is

$$a_{\varepsilon}(\alpha) = p - \kappa_h(\alpha)\varepsilon + o(\varepsilon), \quad \kappa_h(\alpha) = \frac{\delta e^{-mp} \alpha \ell_h(\alpha) \eta_h(\alpha)}{\phi(mp) - \delta e^{-mp} \alpha \rho(p) \ell_h(\alpha)}, \quad (\text{L16})$$

and $\kappa_h(\alpha)$ is strictly increasing in α .

(ii) Fixed price-marginal capacity. Fix $s > 0$ and set

$$\gamma_s(\alpha) = s - \alpha, \quad 0 \leq \alpha \leq \min\{1, s\}. \quad (\text{L17})$$

This holds (L13) fixed at s . With

$$\ell_s(\alpha) = \phi((s - \alpha)mp), \quad \eta_s(\alpha) = \frac{1}{1 - p} \left[1 - \frac{1}{\beta} \left\{ p + \frac{e^{(s - \alpha)mp} - 1}{ms} \right\} \right], \quad (\text{L18})$$

the analogous coefficient

$$\kappa_s(\alpha) = \frac{\delta e^{-mp} \alpha \ell_s(\alpha) \eta_s(\alpha)}{\phi(mp) - \delta e^{-mp} \alpha \rho(p) \ell_s(\alpha)} \quad (\text{L19})$$

is strictly increasing on the active set $\eta_s(\alpha) > 0$.

Thus, within the active local-rescue region, replacing unscreened entrants with screened surviving incumbents strictly increases the first-order acceptance-erosion coefficient under either fixed-supply comparison.

The composition effect on completion is subtler than the cutoff effect.

THEOREM 13 (Closed-form endpoint boundary for local rescue value). Fix $0 < s \leq 1$, impose $\alpha + \gamma = s$, and suppose the all-entrant composition is active:

$$p + \frac{e^{smp} - 1}{ms} < \beta. \quad (\text{L20})$$

Let $L_s(\alpha)$ denote the right derivative in (L5) at composition $(\alpha, s - \alpha)$. Write

$$\begin{aligned} x = mp, \quad R = e^{-x}, \quad E_s = e^{-sx}, \quad \sigma = \phi(x), \quad \rho = \frac{1 - p/\beta}{1 - p}, \\ \eta_E = \frac{1}{1 - p} \left[1 - \frac{1}{\beta} \left\{ p + \frac{e^{sx} - 1}{ms} \right\} \right], \quad H = \frac{E_s \eta_E}{\rho}. \end{aligned} \quad (\text{L21})$$

Then $0 < H < 1$ and the all-entrant and all-incumbent values are

$$L_s(0) = m(1 - p)RsE_s\eta_E, \quad L_s(s) = m(1 - p)Rs\rho \frac{\sigma - \delta R}{\sigma - \delta Rs\rho}. \quad (\text{L22})$$

Define

$$\delta_c(m, p, \beta, s) = \frac{\sigma(1 - H)}{R(1 - Hs\rho)}. \quad (\text{L23})$$

For $\delta \in (0, 1]$,

$$\begin{cases} L_s(s) > L_s(0), & \delta < \delta_c(m, p, \beta, s), \\ L_s(s) = L_s(0), & \delta = \delta_c(m, p, \beta, s), \\ L_s(s) < L_s(0), & \delta > \delta_c(m, p, \beta, s). \end{cases} \quad (\text{L24})$$

Hence the two pure compositions have generically different marginal completion values, but their ranking depends on incumbent patience; $L_s(\alpha)$ need not be monotone between the endpoints.

The nonmonotonicity is not a logical possibility only. At $(m, p, \beta, \delta, s) = (1, 0.1, 0.5, 1, 1)$, direct evaluation of the exact local formula at three strictly mixed compositions gives approximately

$$L_s(0.75) = 0.24267, \quad L_s(0.92) = 0.20589, \quad L_s(0.99) = 0.22333. \quad (\text{L25})$$

Thus local completion value first falls and then rises with the incumbent share even though cut-off erosion rises throughout. This counterexample and the activity boundary are included in the numerical falsification tests.

10. A public-supply benchmark

The Poisson model makes realized local supply latent. To separate that information friction from assignment competition, fix instead a deterministic incumbent count $n \geq 1$ observed by every agent. We evaluate one announced menu conditional on that fixed n ; the exercise removes rival-count uncertainty without introducing state-contingent pricing. Keep the timing, uniform private costs, survival, and no-entry assumption. For a conjectured cutoff a , a focal acceptor faces $X \sim \text{Bin}(n-1, a)$ accepting rivals, so her assignment share is

$$s_n(a) := \mathbb{E} \left[\frac{1}{1+X} \right] = \frac{1 - (1-a)^n}{na}, \quad s_n(0) = 1. \quad (57)$$

Universal rejection occurs against all of her rivals with probability $(1-a)^{n-1}$. Conditional on that event, each rival has cost uniform on $(a, 1)$. If terminal payment p_j is selected, a rival is both surviving and eligible with probability

$$\theta_j(a) = \frac{\alpha(p_j - a)}{1 - a};$$

hence a terminal acceptor's assignment share is $s_n(\theta_j)$ and terminal coverage is

$$C_{j,n}(a) = 1 - [1 - \theta_j(a)]^n = n\theta_j(a)s_n(\theta_j(a)). \quad (58)$$

Let $\eta_j(a)$ be the posting-conditional mass of riders who select terminal payment p_j after failure. At a positive interior cutoff, indifference of the cutoff driver is

$$s_n(a)(p_1 - a) = \delta(1-a)^{n-1} \alpha \sum_j \eta_j(a) s_n(\theta_j(a)) (p_j - a). \quad (59)$$

Substitution of (58)–(59) into unconditional completion gives the exact finite-population counterpart of (20):

$$M_n(p_1, p_2; a) = (1 - p_1) n s_n(a) \left[a + \frac{p_1 - a}{\delta} \right]. \quad (60)$$

PROPOSITION 3 (Public n and local announced rescue). Fix $n \geq 1$, $0 < \alpha \leq 1$, $0 < \delta \leq 1$, and $p \in (0, \beta)$. For every sufficiently small $\varepsilon > 0$, the menu $(p, p + \varepsilon)$ has a unique symmetric equilibrium cutoff, up to behavior of the null cutoff type, and it satisfies

$$a_\varepsilon = p - \kappa_n \varepsilon + o(\varepsilon), \quad \kappa_n = \frac{\delta(1-p)^{n-1}\alpha\rho(p)}{s_n(p) - \delta(1-p)^{n-1}\alpha\rho(p)} > 0. \quad (61)$$

Its right derivative of unconditional completion is

$$\left. \frac{dM_n}{d\varepsilon} \right|_{0+} = (1-p)n\kappa_n \left\{ \frac{1-\delta}{\delta} s_n(p) - p s'_n(p) \right\}. \quad (62)$$

This derivative is strictly positive for every $n \geq 1$ when $\delta < 1$. At $\delta = 1$, it is strictly positive for $n \geq 2$ and exactly zero for $n = 1$.

The denominator in (61) is positive because $s_n(p) \geq (1-p)^{n-1}$ and $\delta\alpha\rho(p) < 1$. Moreover

$$s_n(a) = \frac{1}{n} \sum_{r=0}^{n-1} (1-a)^r,$$

which is strictly decreasing for $n \geq 2$ and constant for $n = 1$. Proposition 3 rules out the claim that an unobserved realized count is necessary for the rescue gain. The first term in (62) is pure intertemporal compensation and survives at $n = 1$; the second is assignment competition and is positive exactly when $n \geq 2$. Latent supply changes posterior inference and the global policy problem, but neither local wedge. If $\alpha = 0$, then $\kappa_n = 0$ and the derivative in (62) is zero for every n .

11. Discussion and Institutional Scope

The fixed- p_1 solution gives a concrete menu-design rule rather than a generic instruction to raise payment after failure. A sufficiently low first payment places the request in a reject-all regime and uses the unique rescue-envelope maximizer. At an intermediate p_1 , the platform chooses a tangent rescue payment that implements the smallest feasible cutoff. Once p_1 reaches rider patience β , a separate rescue payment no longer changes positive-mass continuation behavior.

Three state variables matter operationally. Rider patience determines whether delayed completion remains valuable enough to finance rescue. Market thickness determines how much room the rescue menu has to improve on an already optimized flat payment: $V(m)$ is small when opportunities are scarce and again when flat pricing already completes almost every request. Supply composition matters even at a fixed total expected second-window count. Surviving incumbents have been screened by first-period rejection and discount waiting, whereas fresh entrants are unscreened and respond contemporaneously. Pooling the two groups into a single continuation-supply number therefore removes the very history that drives their different acceptance behavior. The survival

parameter α and the discount factor δ consequently do not conflict: the former changes physical availability and competition, whereas the latter changes only the incumbent's intertemporal payoff.

The public-supply benchmark clarifies two channels. Incumbent impatience can make rescue locally productive even with one known driver; with multiple drivers, assignment competition adds a further incentive to accept early. In very thick markets the optimized rescue policy reduces failure from order $\log m/m$ under flat pricing to order $\log \log m/m$. The absolute completion difference still vanishes, so a large proportional reduction in a small residual failure rate is not a large percentage-point gain.

These implications have a defined institutional scope. The maintained transfer is paid by the rider to the selected driver and the objective is completion. Platform-funded bonuses, profit, budgets, and welfare require additional resource-cost primitives. Simultaneous acceptance with uniform selection represents a broadcast response window, not a first-response race. The equilibrium analysis covers anonymous symmetric independently mixed WPBE, for which cutoff behavior is endogenous; asymmetric and correlated equilibria remain outside the model. Fresh entry is included in the fixed-menu and local-rescue analysis, whereas the global menu and thickness theorems impose $\gamma = 0$. General distributions need not deliver a unique cutoff, and no unique maximizing thickness is asserted even in the uniform core.

Completion, contribution margin, and rider cost. Completion is the appropriate primitive objective when the platform faces a service-level target, an unfilled-request penalty, or a downstream value that is approximately constant across completed requests. It is not a claim that every completion-improving menu raises profit. Let C_t be the event of completion in window t , let r_t be the platform's contribution from such a completion before any platform-funded driver bonus, and let s_t be that bonus. The corresponding accounting objective is

$$\Pi(p_1, p_2) = (r_1 - s_1) \Pr(C_1) + (r_2 - s_2) \Pr(C_2).$$

Our objective sets the two net weights equal and positive. A rescue gain therefore translates directly to profit when payments are rider-funded and per-completion margins are stable, but can reverse when the platform funds a large second-window bonus or delayed service has a much lower margin. Rider cost is not absent from the model: higher payments change posting and continuation through private value v . What remains outside the paper is a normative surplus calculation that assigns social weights to transfers, waiting, and failed service. Figures 3–4 accordingly report completion, not an implicit profit or welfare claim.

When commitment is credible. The strategic-waiting channel requires the complete failure-contingent menu to be visible before the first response window and mechanically honored after failure. Credible implementations include an order card that displays both payments and the trigger, a platform rule that automatically changes the payment after a fixed broadcast window, or a contractual campaign schedule whose second-window payment cannot be revised by a dispatcher. Reputation can also support commitment when drivers repeatedly observe whether announced rescues are honored, but that repeated-game foundation is not modeled here. If the platform chooses p_2 only after observing rejection, or keeps it undisclosed, drivers solve a different game and the announced-menu waiting effect need not arise. This distinction is why the evidence in [Wu et al. \(2022\)](#) is a contrast rather than a direct implementation of the mechanism.

12. Conclusion

An announced rescue menu changes current acceptance as well as continuation coverage. A driver who rejects p_1 may survive to receive p_2 , but waiting also exposes the driver to assignment competition. Universal rejection is therefore both the operational trigger for rescue and a signal about the realized incumbent pool and its costs. Private rider continuation makes the platform solve these supply and demand responses jointly.

With general continuous full-support type distributions, anonymous symmetric independent equilibrium behavior is almost everywhere a pure cutoff, and the cutoff set is nonempty and compact. The economic gain from rescue is not an artifact of uniformity: with no entry, a small rescue increase strictly raises completion around every active flat price under smooth distributions. Convex cost supply yields global same- p_1 dominance branch by branch. The concave and multiple-cutoff examples delimit these conclusions and explain why the unrestricted policy value is evaluated conservatively.

In the uniform no-entry baseline, every menu has a unique equilibrium cutoff. The conditional second-payment problem is a concave envelope, and the global menu problem reduces to a continuous one-dimensional maximum. The optimized rescue menu beats the optimized flat payment above an endogenous rider-patience threshold; with positive incumbent survival, $\beta \geq 1/2$ is sufficient at every finite m . The completion gain $V(m)$ is evaluated at the same thickness under both optimized policies. It vanishes in very thin and very thick markets and reaches a finite positive maximum, without implying a unique peak.

The survival-and-entry extension keeps screening history explicit. Drivers who rejected initially form a truncated incumbent pool and survive with probability α , whereas fresh entrants arrive at intensity γm without that first-period screen. Local rescue remains productive when its activity condition holds. If at least one optimized flat price is active, a small announced rescue strictly

dominates the optimized flat policy even when both sources coexist. This condition is substantive: when it fails strictly, the same small perturbation is exactly unused. Holding either the raw continuation headcount or the price-marginal capacity fixed, a larger incumbent share strictly amplifies first-order strategic waiting within the active local-rescue region. Its marginal completion effect is not monotone in composition; a closed-form endpoint threshold shows how the ranking can reverse with δ . The fully global menu theorem intentionally remains a no-entry result.

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Proofs and Additional Material

EC.1. Proofs for the General Equilibrium and Distributional Results

Proof of Proposition 1. Before any action there is a single response window, so determinism and anonymity assign one initial payment p_1 . Success is terminal. After universal rejection, abandon ends the request, repeat preserves the quoted payment p_1 , and rescue is the only action that activates a distinct payment, denoted p_2 . There is no other public action or message on which the maintained transfer schedule can condition. The restriction that rescue cannot reduce payment is exactly $p_1 \leq p_2$. $\square$

Proof of Lemma 1. Poisson marking splits the incumbent process into costs below and above a , with independent counts of means $mF(a)$ and $m[1 - F(a)]$. Thus first-window coverage is $1 - e^{-mF(a)}$. Under Palm conditioning, a focal driver's accepting rivals form a Poisson random variable Z of mean $mF(a)$, and

$$\mathbb{E}[(1 + Z)^{-1}] = \int_0^1 \mathbb{E}[t^Z] dt = \int_0^1 e^{-mF(a)(1-t)} dt = \phi(mF(a)).$$

If the focal driver waits, continuation is reached only if every rival has cost above a , an event of probability $e^{-mF(a)}$. Conditional on no point below a , the above- a process is unchanged. Independent survival thins the eligible interval $(a, p_j]$ to intensity $m\alpha[F(p_j) - F(a)]$; fresh eligible drivers contribute $m\gamma F(p_j)$. Superposition gives (2), and the same integral identity gives terminal assignment share $\phi(\lambda_j)$. $\square$

Proof of Theorem 1. For an arbitrary symmetric mixed strategy, equations (5)–(6) give the accept-versus-wait margin for every $c \leq p_1$. If $q > 0$,

$$B(\sigma) \geq \phi(mq) - e^{-mq} > 0.$$

If $q = 0$, then

$$B(\sigma) \geq 1 - \delta\alpha\rho_G(p_1) > 0,$$

because $p_1 > 0$, $\beta < 1$, and full support imply $\rho_G(p_1) < 1$. The margin is therefore strictly decreasing in cost. Drivers with $c > p_1$ cannot profitably accept immediately. Hence the accepting set is an interval and mixing can occur only at its boundary. Atomlessness makes that type irrelevant for outcomes. Conditioning on a public sunspot realization leaves the same argument unchanged. $\square$

Proof of Theorem 2. Given a posted request, failure is independent of v , so the rider retains the distribution G truncated to $[p_1, 1]$. Comparing the affine continuation payoffs gives v^M and (3). Before posting, a type $v < p_1$ cannot obtain positive current or delayed surplus. A type $v \geq p_1$ can

abandon after failure and, under the stated zero-payoff rule, posts. Thus the posting set is exactly $[p_1, 1]$.

Theorem 1 reduces equilibrium to a numerical cutoff. Evaluation of (6) at that type gives (7). At $a = p_1$, $f_{F,G}(p_1) \leq 0$. If $f_{F,G}(0) \leq 0$, the lower boundary is an equilibrium. If $f_{F,G}(0) > 0$, either the upper margin is zero or the intermediate value theorem supplies an interior zero. The rider's aggregate continuation payoff is continuous at an action switch because the two actions tie there; when continuation coverage vanishes, waiting payoff vanishes. Hence the residual is continuous. The boundary inequalities and zero set are closed, so the nonempty equilibrium set is compact. Formula (9) is continuous on that set, which makes (10) well defined. $\square$

Proof of Proposition 2. For $a < p$, the cutoff margin is $p - a$ times the bracket displayed below (13). If $F(a) > 0$, then $\phi(mF(a)) > e^{-mF(a)}$ and the remaining multiplier is at most one. At $a = 0$, strictness follows from $\rho_G(p) < 1$. No cutoff below p is sustainable, while $a = p$ is the unique equilibrium cutoff. At that cutoff, no surviving incumbent can accept p after rejection. Initial completion has mass $[1 - G(p)](1 - e^{-mF(p)})$; after failure, rider mass $1 - \tilde{G}(p/\beta)$ can be served by fresh entrants with coverage $1 - e^{-\gamma mF(p)}$. This gives (12). Continuity on $[0, 1]$ gives attainment, and differentiating (F3) gives (F4). $\square$

Proof of Theorem 3. Let $d_\varepsilon = p - a_\varepsilon$. Cutoff incentives imply

$$[\phi(mF(a_\varepsilon)) - \delta\alpha e^{-mF(a_\varepsilon)}\rho_G(p)]d_\varepsilon \leq \delta\alpha e^{-mF(a_\varepsilon)}\rho_G(p)\varepsilon. \quad (\text{EC.G1})$$

The bracket is uniformly bounded away from zero, so every equilibrium has $d_\varepsilon = O(\varepsilon)$. This excludes the lower boundary, and the upper boundary is not an equilibrium because a positive rider mass strictly prefers rescue. Every equilibrium is therefore interior and localizes at p .

Differentiability of F gives

$$F(p) - F(a_\varepsilon) = f(p)d_\varepsilon + o(\varepsilon), \quad F(p + \varepsilon) - F(p) = f(p)\varepsilon + o(\varepsilon).$$

Uniformly on the compact rescaling $d_\varepsilon/\varepsilon$, the repeat mass converges to zero, the rescue mass converges to $\rho_G(p)$, and both terminal assignment shares converge to one. Dividing cutoff indifference by ε yields

$$\sigma\kappa = \delta\alpha R\rho_G(p)(1 + \kappa),$$

whose positive slope gives the common expansion (G1) for every branch.

Relative to flat pricing, the first-window coverage derivative is $-mf(p)R\kappa$, while terminal rescue contributes $mf(p)R\alpha\rho_G(p)(1 + \kappa)$. Multiplication by posting mass $1 - G(p)$ and substitution of (G1) give (G2). Positivity follows from $\sigma > \delta R$. The expansion is uniform across the compact equilibrium set, so taking the minimum preserves its strictly positive first-order term. Applying the result at an optimal flat price below β proves the final claim. $\square$

Proof of Theorem 4. Let

$$x_j = F(p_j) - F(a), \quad s_j = \frac{x_j}{p_j - a}, \quad K = \eta^R C_1 + \eta^2 C_2.$$

Convexity gives $s_2 \geq s_1 > 0$. At an interior cutoff, immediate acceptance and waiting are equal; at the lower-boundary cutoff, waiting is weakly preferred. Using $C_j = m\alpha x_j \phi(m\alpha x_j)$ therefore gives

$$\phi(mF(a))(p_1 - a) \leq \frac{\delta e^{-mF(a)}}{m} \left(\frac{\eta^R C_1}{s_1} + \frac{\eta^2 C_2}{s_2} \right) \leq \frac{\delta e^{-mF(a)}}{ms_1} K. \quad (\text{EC.G2})$$

For $a > 0$, (EC.G2) implies

$$K \geq \frac{x_1}{\delta F(a)} \{e^{mF(a)} - 1\} > mx_1.$$

At $a = 0$, the limiting inequality gives $K \geq mx_1/\delta$. In either case $K > 1 - e^{-mx_1}$. Consequently

$$\begin{aligned} M_{F,G}^{\alpha,0}(p_1, p_2; a) - [1 - G(p_1)]\{1 - e^{-mF(p_1)}\} \\ = [1 - G(p_1)]e^{-mF(a)}\{K - (1 - e^{-mx_1})\} > 0, \end{aligned}$$

which proves (G3). $\square$

EC.2. Distributional Counterexamples and Equilibrium Selection

The concave-CDF example in Section 6 is obtained by substituting $F(c) = \sqrt{c}$ into (7). High-precision evaluation gives

$$f(0.0203133) = 6.2095 \times 10^{-8} > 0, \quad f(0.0203134) = -3.8965 \times 10^{-8} < 0.$$

The intermediate value theorem therefore places an equilibrium cutoff in that interval. Direct interval evaluation bounds its completion between 0.1034830 and 0.1034833, while (F3) gives 0.1083552320. Solving within the bracket gives $a = 0.0203133614$ and completion 0.1034831363. Thus the flat policy's advantage exceeds 0.0048.

For the multiplicity example, define

$$H_{[\ell,u]}(c) = \begin{cases} 0, & c \leq \ell, \\ (c - \ell)/(u - \ell), & \ell < c < u, \\ 1, & c \geq u. \end{cases}$$

Set

$$F(c) = 0.2c + 0.4H_{[0.04,0.045]}(c) + 0.4H_{[0.15,0.155]}(c), \quad G(v) = v^{20},$$

and

$$(m, \alpha, \beta, \delta, \gamma; p_1, p_2) = (2, 0.7, 0.9, 1, 0; 0.1, 0.2).$$

The residual (7) has at least three interior zeros in the disjoint brackets

bracket	$f(\text{left})$	$f(\text{right})$
$[0.03035, 0.03037]$	5.8570×10^{-6}	-5.7408×10^{-6}
$[0.04343, 0.04345]$	-1.3174×10^{-5}	1.0589×10^{-5}
$[0.04859, 0.04862]$	6.3290×10^{-6}	-7.0902×10^{-6} .

Their numerical locations are

$$a_1 = 0.0303601002, \quad a_2 = 0.0434410792, \quad a_3 = 0.0486041490,$$

with slopes approximately -0.580 , 1.188 , and -0.447 . Their completion probabilities are 0.6926116 , 0.7398208 , and 0.7587285 ; the first and third differ by more than 0.066 .

Let $q = (p_1, p_2)$ range over a compact menu set bounded away from the null posting boundary, and let $\mathcal{E}(q)$ be the cutoff set. Define optimistic and pessimistic branch values

$$O(q) = \max_{a \in \mathcal{E}(q)} M(q, a), \quad P(q) = \min_{a \in \mathcal{E}(q)} M(q, a).$$

Selection topology. If $\mathcal{E}$ is nonempty, compact valued, and has a closed graph, then it is upper hemicontinuous, O is upper semicontinuous, and P is lower semicontinuous. The optimistic design problem attains a maximum on a compact menu set; the pessimistic supremum need not be attained. If $\mathcal{E}$ is also lower hemicontinuous, both values are continuous and both maxima are attained.

Proof of the selection-topology statement. For any $q_n \rightarrow q$, compactness gives convergent subsequences of equilibrium selections, and the closed graph puts every limit in $\mathcal{E}(q)$. Applying this observation to maximizing selections gives $\limsup O(q_n) \leq O(q)$; applying it to minimizing selections gives $P(q) \leq \liminf P(q_n)$. Upper semicontinuity on a compact set yields optimistic attainment. Lower semicontinuity alone does not ensure a maximum: for $\mathcal{E}(q) = \{q\}$ when $q < 1$, $\mathcal{E}(1) = \{0, 1\}$, and $M(q, a) = a$, one has $P(q) = q$ for $q < 1$ but $P(1) = 0$. Lower hemicontinuity and the maximum theorem give continuity and attainment for both selections. $\square$

Proof of Lemma 2. For $p < \beta$,

$$M_{\gamma}^{\text{flat}}(m, p) = q_0(m, p) + e^{-mp}(1 - p/\beta)(1 - e^{-\gamma mp}), \quad (\text{EC.1})$$

whereas $M_{\gamma}^{\text{flat}}(m, p) = q_0(m, p)$ for $p \geq \beta$. The one-sided derivatives at the rider-continuation kink satisfy

$$M_{\gamma,+}^{\text{flat}'}(m, \beta) - M_{\gamma,-}^{\text{flat}'}(m, \beta) = \frac{e^{-m\beta}(1 - e^{-\gamma m\beta})}{\beta} > 0. \quad (\text{EC.2})$$

A local maximum requires a nonnegative left derivative and a nonpositive right derivative, so this upward jump excludes $p = \beta$. Moreover, $M_{\gamma,+}^{\text{flat}'}(m, 0) = m(1 + \gamma) > 0$, and the endpoint $p = 1$ has zero completion, so neither endpoint is optimal.

Direct differentiation gives

$$q_{0,pp}(m, p) = -me^{-mp}\{2 + m(1 - p)\} < 0. \quad (\text{EC.3})$$

Thus q_0 is strictly concave, and its first-order condition is $e^{mp} = 1 + m(1 - p)$. Solving it gives $p^0(m)$ in (F1). The left side minus the right side is negative at zero and strictly positive at $p = 1/2$

because $e^{m/2} > 1 + m/2$; hence $0 < p^0(m) < 1/2$. On $[\beta, 1]$ the only possible global maximizer is therefore $p^0(m)$, and it lies in that interval only if $\beta < p^0(m)$. This proves (F2).

It remains to verify the stated nonuniqueness possibility. Fix $\beta < p^0(m)$ and extend the objective continuously to $\gamma = 0$. Let

$$B(\gamma) = \max_{p \in [0, \beta]} M_\gamma^{\text{flat}}(m, p).$$

The maximum theorem makes B continuous. At zero, $B(0) = q_0(m, \beta) < q_0(m, p^0)$. For $p_\gamma = (\log \gamma)/(\gamma m)$, which lies below β for large γ , equation (EC.1) gives $M_\gamma^{\text{flat}}(m, p_\gamma) \rightarrow 1 > q_0(m, p^0)$. The intermediate value theorem therefore gives some $\gamma^\dagger > 0$ with $B(\gamma^\dagger) = q_0(m, p^0)$. At this value p^0 and a point strictly below β are both global flat maximizers; the latter cannot equal β because $q_0(m, \beta) < q_0(m, p^0)$. $\square$

Proof of Theorem 10. Put $d = p - a$. Under $(p, p + \varepsilon)$ the two terminal intensities are

$$\lambda_1 = m(\alpha d + \gamma p), \quad \lambda_2 = \lambda_1 + m(\alpha + \gamma)\varepsilon. \quad (\text{A1})$$

If $\eta_\varepsilon(a)$ is the rescue mass, interior cutoff indifference is

$$\begin{aligned} \phi(ma)d &= \delta \alpha e^{-ma} \{ [\rho - \eta_\varepsilon] \phi(\lambda_1) d \\ &\quad + \eta_\varepsilon \phi(\lambda_2) (d + \varepsilon) \}. \end{aligned} \quad (\text{A2})$$

Rearrange and use $\phi(\lambda_j) \leq 1$, $\eta_\varepsilon \leq \rho$, and $\phi(ma) \geq e^{-ma}$. The coefficient on d is at least $e^{-ma}(1 - \delta \alpha \rho) > 0$, which gives (L4). The same weak boundary inequality excludes a reject-all cutoff for small ε ; the upper cutoff already obeys the bound. Thus every equilibrium is localized, not only a selected branch.

On the compact rescaling $y = d/\varepsilon$, (A1) gives

$$\frac{C_2 - C_1}{\varepsilon} \rightarrow mE(\alpha + \gamma), \quad C_2 \rightarrow 1 - E.$$

The rider switch therefore converges uniformly to $\bar{v}$ in (L2), and the rescue mass converges to $\eta^0 > 0$. Divide (A2) by ε and pass to the limit:

$$[\sigma - \delta R \alpha \rho \ell] y - \delta R \alpha \eta^0 \ell = 0. \quad (\text{A3})$$

Its slope is at least $\sigma - R > 0$. The rescaled margin converges in C^1 to this affine function, so it has a unique zero for all small ε ; (A3) yields (L3).

It remains to establish the strict sign in (L6). First set $\delta = 1$ and call the resulting expression T_1 . The case $\gamma = 0$ gives $T_1 = \alpha(\sigma - R) > 0$. For $\gamma > 0$, set

$$u = e^x - 1, \quad t = 1 - E, \quad A = \alpha + \gamma, \quad \Delta = 1 - E(1 + \gamma), \quad B = 1 - \rho \Delta.$$

Since $\sigma = Ru/x$ and $\ell = t/(\gamma x)$,

$$T_1 = \frac{R}{\gamma x} Q, \quad Q = E\gamma Au - \alpha t B. \quad (\text{A4})$$

If $\Delta \leq 0$, then $B \leq E(1 + \gamma)$, $\gamma u > t$, and $A \geq \alpha(1 + \gamma)$, so $Q > 0$. If $\Delta > 0$ and $\gamma \leq 1$, convexity gives $t \leq E\gamma u$ and $B \leq 1$, whence $Q \geq \alpha(E\gamma u - t) + E\gamma^2 u > 0$.

Finally suppose $\Delta > 0$ and $\gamma > 1$. Activity implies the weaker strict bound

$$\rho > \frac{t}{t + xEA}.$$

It implies

$$Q \geq \frac{E}{t + xEA} I(\alpha), \quad I(\alpha) = \gamma Au(t + xEA) - \alpha t \{xA + t(1 + \gamma)\}. \quad (\text{A5})$$

The coefficient on α^2 in I is $x(\gamma Eu - t) < 0$, because $e^{\gamma x} - 1 > \gamma(e^x - 1)$. Hence the minimum of this concave quadratic on $[0, 1]$ is at an endpoint. Clearly $I(0) > 0$. At $\alpha = 1$, write $H = t + xE(1 + \gamma)$ and $d = \gamma u - t$. Then

$$\frac{I(1)}{1 + \gamma} = dH - tx\Delta. \quad (\text{A6})$$

Let $w = \gamma x$. Since $\Delta > 0$, $w > \log(1 + \gamma)$; moreover

$$d > \frac{\gamma x^2}{2} = \frac{xw}{2}, \quad \Delta = 1 - (1 + \gamma)e^{-w} < \frac{w}{2}.$$

The last inequality follows by minimizing $w/2 - 1 + (1 + \gamma)e^{-w}$ on $[\log(1 + \gamma), \infty)$. Thus $d > x\Delta$ and, since $H > t$, (A6) is positive. This proves $Q > 0$ in all cases and hence $T_1 > 0$. For general $0 < \delta \leq 1$,

$$T_\delta = T_1 + (1 - \delta)R\alpha\ell\{1 - \rho + \rho E(1 + \gamma)\} > 0.$$

Along the cutoff branch, write continuation coverage as $S_\varepsilon = \rho C_1 + \eta_\varepsilon(C_2 - C_1)$. Direct differentiation at zero gives

$$S'_0 = mE\{\rho\alpha\kappa + \eta^0(\alpha + \gamma)\}.$$

Differentiating (9), substituting (L3), and collecting terms yields exactly (L5)–(L6). All remaining factors are positive, completing the proof. $\square$

Proof of Theorem 11. Condition (L9) is exactly the activity condition $\bar{v} < 1$ in Theorem 10, evaluated at $p = p^*$. That theorem gives a unique cutoff and the expansion

$$M_m^{\alpha, \gamma}(p^*, p^* + \varepsilon; a_\varepsilon) = M_m^{\alpha, \gamma}(p^*, p^*; p^*) + \Lambda(p^*)\varepsilon + o(\varepsilon), \quad (\text{EC.4})$$

where $\Lambda(p^*)$ is the strictly positive expression in (L5). Since $p^* \in \mathcal{P}_\gamma^{\text{flat}}(m)$, the first term on the right is $M_\gamma^{\text{flat}, *}(m)$. Uniqueness makes the conservative minimum in (10) equal to this branch value. A sufficiently small positive ε therefore strictly improves on $M_\gamma^{\text{flat}, *}(m)$, and the supremum in (11)

is at least that strictly higher value. This proves (L10)–(L11) without requiring the supremum to be attained.

Now suppose $\mathcal{A}_{\alpha,\gamma}(p) < 0$. The localization argument leading to (L4) uses only $\eta_\varepsilon \leq \rho$ and hence still applies. On its compact rescaling $0 \leq (p-a)/\varepsilon \leq \delta\alpha\rho/(1-\delta\alpha\rho)$, the repeat-rescue switch converges uniformly to $\bar{v} > 1$. Thus $\eta_\varepsilon = 0$ throughout that set for all small ε . For every $a < p$, cutoff indifference then has the strict flat-sign factor

$$(p-a) [\phi(ma) - \delta\alpha e^{-ma} \rho(p) \phi(m\{\alpha(p-a) + \gamma p\})] > 0. \quad (\text{EC.5})$$

Consequently no cutoff below p is sustainable, while $a = p$ is eligible and unique. Rescue is unused and the exact completion is $M_\gamma^{\text{flat}}(m, p)$. At equality the limiting rescue mass is zero and this first-order argument does not determine the sign. $\square$

Proof of Theorem 12. Write $x = mp$, $R = e^{-x}$, and $\sigma = \phi(x)$. The local expansion (L3) can be written for any active composition as

$$\kappa = \frac{\delta R q \eta}{\sigma - \delta R \rho(p) q}, \quad q = \alpha \phi(\gamma x). \quad (\text{EC.6})$$

The denominator is strictly positive because $q \leq \alpha \leq 1$, $\rho(p) < 1$, and $\sigma > R$.

Along the fixed-headcount path, $\gamma'_h(\alpha) = -(1-p)$ and $A'_h(\alpha) = p$. Therefore

$$\frac{d}{d\alpha} \frac{e^{\gamma_h(\alpha)x} - 1}{A_h(\alpha)} = \frac{-(1-p)x e^{\gamma_h(\alpha)x} A_h(\alpha) - p\{e^{\gamma_h(\alpha)x} - 1\}}{A_h(\alpha)^2} < 0. \quad (\text{EC.7})$$

It follows that η_h is strictly increasing. Moreover,

$$\phi'(z) = \frac{e^{-z}(1+z) - 1}{z^2} < 0 \quad (z > 0), \quad (\text{EC.8})$$

so the decrease in γ_h makes ℓ_h increase and makes $q_h(\alpha) = \alpha \ell_h(\alpha)$ strictly increase. For positive q and η , the right side of (EC.6) satisfies

$$\frac{\partial \kappa}{\partial q} = \frac{\delta R \eta \sigma}{\{\sigma - \delta R \rho(p) q\}^2} > 0, \quad \frac{\partial \kappa}{\partial \eta} = \frac{\delta R q}{\sigma - \delta R \rho(p) q} > 0. \quad (\text{EC.9})$$

This proves part (i), including comparisons from $\alpha = 0$ by continuity.

Along the fixed-capacity path, $\gamma'_s(\alpha) = -1$ while $\alpha + \gamma_s(\alpha) = s$. Hence

$$\frac{d}{d\alpha} \frac{e^{\gamma_s(\alpha)x} - 1}{s} = -\frac{x e^{\gamma_s(\alpha)x}}{s} < 0. \quad (\text{EC.10})$$

Thus η_s , ℓ_s , and $\alpha \ell_s$ all increase strictly on the active set. Equation (EC.9) proves part (ii). $\square$

Proof of Theorem 13. Condition (L20) is $\eta_s(0) > 0$. Since $\eta_s(\alpha)$ increases in α , the full path is active. At the all-entrant endpoint $(\alpha, \gamma) = (0, s)$, cutoff erosion is zero and substitution into (L5)–(L6) gives the first expression in (L22). At the all-incumbent endpoint $(\alpha, \gamma) = (s, 0)$, we have $E = \ell = 1$ and $\eta^0 = \rho$, so the same formula gives the second expression in (L22).

Positive entry raises the repeat–rescue switch strictly above p/β . Thus (L20) implies $0 < \eta_E < \rho$, and $0 < H < 1$. Also $s\rho < 1$, so every denominator below is positive. Removing the common positive factor $m(1-p)Rs\rho$ from $L_s(s) - L_s(0)$ leaves the sign of

$$\frac{\sigma - \delta R}{\sigma - \delta Rs\rho} - H = \frac{\sigma(1-H) - \delta R(1-Hs\rho)}{\sigma - \delta Rs\rho}. \quad (\text{EC.11})$$

The numerator crosses zero exactly at (L23), proving (L24). This comparison ranks only the two pure endpoints. The values in (L25), obtained from the same exact local formula at an interior composition, establish that the path itself need not be monotone. $\square$

EC.3. Proofs for global no-entry design

Proof of Theorem 5. The cancellation leading to (16) follows by substituting the rider switch type into the sum of repeat and rescue coverage. Inequality (18) excludes the repeat branch and its kink from the zero set. Every zero is therefore a smooth active-rescue zero. With the notation preceding (19), its root equation, $B - b = p_2 - \beta p_1 > 0$, and $\delta \leq 1$ imply $u < C(z)/m$. Consequently

$$\frac{1}{u} - \frac{k}{e^{kz} - 1} > m \frac{1 - \alpha e^{-kz}}{1 - e^{-kz}} \geq m > \frac{h'(a)}{h(a)},$$

where $h'(a)/h(a) < m$ follows from $h(a) = \int_0^m e^{ta} dt$. Thus (19) is strictly negative and every zero crosses downward. Also $\mathcal{D}_{12}^\delta(p_1) = -\delta(\beta - p_2)C(p_2 - p_1)/B_1 < 0$. A continuous function whose every zero is a strict downward crossing has at most one zero. The endpoint signs give exactly the two cases stated in the theorem, including the equality case at zero. Flat and repeat-only menus use the strict flat inequality and have cutoff p_1 ; if $p_1 \geq \beta$, no positive mass continues; if $\alpha = 0$, waiting has zero payoff; and $p_1 = 0$ has only the lower numeric cutoff. These observations cover every no-entry menu. $\square$

Proof of Theorem 6. For a target cutoff a , active-rescue implementability is exactly (31). The strict flat inequality makes its right side larger than $A(a, p_1)$, so an implementer is strictly inside the rescue region. The function $A(a, p_2)$ is strictly concave because

$$A_{p_2 p_2}(a, p_2) = -e^{-k(p_2 - a)}[2k + k^2(\beta - p_2)] < 0,$$

and its unique maximizer is $Q(a)$. At $a = 0$, reject-all is implementable iff $A(0, p_2) \geq \beta(1 - p_1)mp_1/\delta$. Its largest possible left side is S_0 , so the condition is feasible exactly when $p_1 \leq p_z$; the strict flat inequality at $(a, p_1) = (0, Q_0)$ gives $0 < p_z < Q_0 < 1/2$.

If $p_1 \leq p_z$, any positive-cutoff completion is below $(1 - p_1)mp_1/\delta \leq S_0/\beta$, whereas reject-all completion is $A(0, p_2)/\beta \leq S_0/\beta$. Strict concavity makes $p_2 = Q_0$ the unique maximizer, including at $p_1 = p_z$. If $p_z < p_1 < \beta$, maximize A pointwise in p_2 and define the envelope margin

$$\overline{\mathcal{D}}_{p_1}(a) = (p_1 - a)h(a) - \delta S(a)/[\beta(1 - p_1)].$$

It is positive at zero and negative at p_1 . The same logarithmic derivative argument as in Theorem 5 shows that every envelope zero crosses downward, hence (30) has exactly one root $a^*(p_1)$. Any menu implementing a satisfies $A(a, p_2) \leq S(a)$, so its unique cutoff cannot lie below $a^*(p_1)$. Completion in (20) decreases with a positive cutoff; therefore the tangent menu $p_2 = Q(a^*)$ is the unique maximizer. For $p_1 \geq \beta$, no positive mass uses rescue and all $p_2 \geq p_1$ are outcome-equivalent to the flat payment. This proves all three regimes and attainment. $\square$

Proof of Theorem 7. Let $g_a(p_1) = (1 - p_1)(p_1 - a)$. Equation (26) implies $Q(a) < (1 + a)/2$, so g_a is strictly increasing on $(a, Q(a))$. The strict flat inequality gives $g_a(Q(a)) > R_\delta(a)$, while $g_a(a) = 0 < R_\delta(a)$; hence (32) has exactly one root in that interval, the lower branch (33). If two distinct cutoffs had the same value of P_δ , the same fixed- p_1 envelope would have two downward-crossing zeros. This also excludes $P_\delta(a) = P_\delta(0)$ for $a > 0$. Thus the continuous map P_δ is strictly increasing from $[0, \beta]$ onto $[p_z, \beta]$.

The three fixed-price regimes now give exactly (35). Both objectives are continuous on fixed compact intervals, so each maximum and the original policy maximum are attained. The flat objective is strictly concave; its first-order condition is $e^{mp_F} - 1 = m(1 - p_F)$, whose unique root satisfies $p_F < 1/2$. For any $p_1 < p_2 < \beta$, Theorem 5 gives a cutoff below p_1 ; equations (20)–(21) then make completion strictly larger than $F_m(p_1)$. If $\beta \geq 1/2$, apply this comparison at $p_F < \beta$. The reject-all boundary and $p_1 \geq \beta$ flat region do not beat F_0^* , whereas the strict comparison does, so every global maximizer is a strict interior menu parameterized as in (36). No step requires uniqueness of the outer argmax. $\square$

Proof of Theorem 8. First fix a strict menu $p_1 < p_2$. If $\beta \leq p_2$, rescue has no positive-mass use and the strict flat inequality forces cutoff p_1 , so the menu value is $F_m(p_1)$. If $p_2 < \beta_1 < \beta_2$, the coefficients $(\beta - z)/[\beta(1 - p_1)]$, $z \in \{p_1, p_2\}$, weakly increase in β and the active rescue coefficient increases strictly. At the β_1 cutoff this strictly lowers the scalar margin for β_2 . Global downward crossing therefore moves the β_2 cutoff strictly left, unless it is already zero. Equations (20)–(21) then strictly increase completion. At a zero cutoff the value is $(1 - p_2/\beta)C(p_2)$ and has positive derivative. As $\beta \downarrow p_2$, any cutoff limit below p_1 would contradict (18), so the menu value is continuous at p_2 .

Continuity of the optimized value follows from the fixed coordinate $a = \beta x$, $x \in [0, 1]$. The unique solution of $e^{ky} - 1 = k\{\beta(1 - x) - y\}$ and all objects $Q, S, P_{\beta, \delta}, J_{\beta, \delta}$ are jointly continuous, with

the removable extensions described before (39). The restricted flat term is jointly continuous after writing $p = \beta + (1 - \beta)s$. The maximum theorem applied to these fixed compact sets makes $D(m; \beta)$ continuous.

Flat policies give $D \geq F_0^*$. If strict gain holds at β_1 , attainment supplies an optimal strict menu with $p_2 < \beta_1$. Reusing it at any $\beta_2 > \beta_1$ and the preceding fixed-menu result gives strict gain and a strictly larger optimized value. Thus the gain set is an open upper interval and has the form (40). An active menu has $p_2 < \beta$ and completion at most $1 - e^{-mp_2} < 1 - e^{-m\beta}$; this proves the positive lower bound in (41). If $\beta > p_F$, a strict menu at first payment p_F beats the flat optimum by (22), proving the upper bound. Finally, the second maximum in (35) never exceeds F_0^* , so strict gain is equivalent to the scalar test (42). If $\alpha = 0$, no incumbent can complete after waiting and the dynamic and flat values coincide. $\square$

EC.4. Proofs for thickness and the public-supply benchmark

Proof of Theorem 9. Equations (51)–(53) are the fixed-coordinate form of (26)–(34). The inverse of $z \mapsto z + \log(1 + z)$ is continuous. Since multiplying the old envelope term by $0 < \delta \leq 1$ only enlarges the lower-root discriminant, the strict envelope bound keeps that discriminant positive. Thus J_m^δ is jointly continuous on every compact m -interval times $[0, \beta]$. Berge's maximum theorem applied to (35) proves part (i).

Uniform thin expansion and localization. Put $\bar{J}_m^\delta(a) = J_m^\delta(a)/m$ and $\bar{J}_0^\delta(a) = H_0^\delta(a)$. The maps $y_m, T_m/m, B_m^\delta, P_m^\delta$, and $\bar{J}_m^\delta$ have jointly analytic removable extensions to a neighborhood of $[0, \bar{m}] \times [0, \beta]$ for some $\bar{m} > 0$. Indeed, $z \mapsto z + \log(1 + z)$ has derivative two at zero, while

$$B_m^\delta(a) = \delta \frac{ma}{e^{ma} - 1} \frac{\alpha y_m(a)^2}{\beta[1 + \alpha m y_m(a)]}.$$

The limiting square-root discriminant is uniformly separated from zero:

$$(1 - a)^2 - \frac{\delta\alpha}{\beta}(\beta - a)^2 \geq (1 - \beta)(1 - a)^2 \geq (1 - \beta)^3 > 0.$$

Thus every finite-order Taylor expansion used below has a remainder uniform in $a \in [0, \beta]$. In particular,

$$y_m(a) = \frac{\beta - a}{2} + O(m), \quad \frac{S_m(a)}{m} = \frac{\alpha(\beta - a)^2}{4} + O(m),$$

and $\bar{J}_m^\delta \rightarrow H_0^\delta$ uniformly.

If $\beta > 1/2$ and $\delta < 1$, put $r(a) = \delta\alpha(\beta - a)^2 - \beta(1 - 2a)$. Then $r(0) < 0$, $r(1/2) > 0$, and $r'(a) = 2\{\beta - \delta\alpha(\beta - a)\} > 0$. Hence a unique $a \in (0, 1/2)$ satisfies $P_0^\delta(a) = 1/2$. The value displayed after (53) is strictly above $1/4$, so uniform convergence and $F_0^*(m)/m \rightarrow 1/4$ prove (45).

If $\delta = 1$ and $\beta > 1/2$, then

$$\bar{J}_m^1(a) = H_0^1(a) + mh_1(a) + o(m)$$

uniformly. The unique maximizer of $H_0^1 = P_0^1(1 - P_0^1)$ is a_0 , and direct expansion at $P_0^1(a_0) = 1/2$ gives $h_1(a_0) = -a_0/8$. The uniform argmax lemma therefore yields

$$\max_a \bar{J}_m^1(a) = \frac{1}{4} - \frac{a_0}{8}m + o(m).$$

For completeness, that lemma follows because every maximizing sequence converges to the unique maximizer of the uniform limit; evaluation at a_0 gives the lower bound, while $H_0^1(a_m) \leq H_0^1(a_0)$ and continuity of h_1 give the upper bound. Together with $F_0^*(m) = m/4 - m^2/16 + o(m^2)$, this proves (46).

Now let $\beta = 1/2$, write $d = 1/2 - a$, and put $A_\delta = 1 - \alpha(1 - \delta)/2$. As $d \downarrow 0$,

$$H_0^\delta(1/2) - H_0^\delta(1/2 - d) = A_\delta d^2 + O(d^3).$$

Uniform convergence first implies $d_m \rightarrow 0$ for every maximizing sequence. Writing $\bar{J}_m^\delta = H_0^\delta + mh_1^\delta + O(m^2)$ uniformly and using local Lipschitz continuity of h_1^δ , comparison with the feasible endpoint gives constants independent of the sequence such that

$$c_1 d_m^2 \leq c_2 m d_m + c_3 m^2.$$

Hence $d_m = O(m)$, so $a_m = 1/2 - c_m m$ with c_m in one fixed compact interval. On that interval (51)–(53) give (54) uniformly. For $\delta < 1$, maximizing its strict quadratic at $c_\delta = 1/(16A_\delta)$ and subtracting the flat expansion proves (47). At $\delta = 1$, let $c_0 = 1/16$. If L is a Lipschitz constant for f_3 on the localization interval, then

$$f_2(c) + mf_3(c) \leq f_2(c_0) + mf_3(c_0) - (c - c_0)^2 + mL|c - c_0|.$$

The displacement terms have supremum $O(m^2)$, while evaluation at c_0 gives the matching lower bound. Thus (54a)–(54b) prove (48) without postulating a higher-order expansion of the optimizer.

If $\beta < 1/2$, $P_0^\delta(a) \geq a$ and $\alpha \leq 1$ imply the bound stated after (54). Its right side is increasing on $[0, \beta]$ and ends at $\beta(1 - \beta) < 1/4$. Uniform convergence and $p_F(m) \rightarrow 1/2 > \beta$ make the flat term the exact winner for all sufficiently small m , proving (49).

Uniform thick-market bound. Put $x = ma$, $q = Q_m(a)$, and $y = \beta - q$. Equations (51) imply

$$q - a = \frac{\log(1 + \alpha my)}{\alpha m}, \quad 1 - T_m(a) = \frac{q}{\beta} + \frac{y}{\beta(1 + \alpha my)}.$$

For the feasible cutoff $x_m = \log \log m$, the first identity gives $q = O(\log m/m) = o(1)$, so $y \geq \beta/2$ eventually and $1 - T_m = O(\log m/m)$. Since the tangent lower root satisfies $p_1 \leq q$, its implementation equation gives, uniformly over $0 < \delta \leq 1$,

$$0 \leq p_1 - a \leq \frac{2a}{e^{x_m} - 1} = O\left(\frac{a}{\log m}\right).$$

Substitution into (55) gives

$$1 - M_R^*(m) \leq (1 + o(1)) \frac{\log \log m}{m}.$$

For the matching policy-uniform lower bound, first suppose $q \leq \beta/2$. Then $y \geq \beta/2$, $p_1 \geq a = x/m$, and the exact identities imply

$$m[1 - J_m^\delta(a)] \geq x(1 - e^{-x}) + C_m e^{-x}, \quad C_m = \frac{\log(1 + \alpha m \beta/2)}{\alpha \beta}.$$

For $C \rightarrow \infty$, $\inf_{x \geq 0} \{x(1 - e^{-x}) + C e^{-x}\} \sim \log C$: evaluation at $x = \log C$ supplies the upper bound, and splitting at $x = (1 - \varepsilon) \log C$ supplies the lower bound. Since $\log C_m \sim \log \log m$, this proves the required rate on this branch. If instead $q > \beta/2$, then

$$a = q - \frac{\log(1 + \alpha m y)}{\alpha m} > \frac{\beta}{2} - \frac{\log(1 + \alpha m \beta)}{\alpha m} > \frac{\beta}{4}$$

eventually. Since $P_m^\delta(a) \geq a$, (55) gives

$$1 - J_m^\delta(a) \geq \frac{\beta}{4}(1 - e^{-m\beta/4}),$$

which dominates $\log \log m/m$. The separate flat branch $p \geq \beta$ has loss at least β . Thus the lower bound is uniform over the full policy domain, proving the first relation in (50).

Finally, if $x_F = mp_F(m)$, the flat first-order condition is $e^{x_F} = 1 + m - x_F$ and gives the exact identity

$$1 - M_F^*(m) = \frac{x_F + 1 - e^{-x_F}}{m}.$$

Since $x_F \sim \log m$, the second relation in (50) follows, and subtracting the two losses gives the third. The dynamic lower bound is independent of δ , while the feasible upper bound used only $\delta \leq 1$; this proves the stated uniformity and its primitive-boundary qualification. $\square$

Proof of Corollary 1. The thin regimes imply $V(m) \rightarrow 0$ as $m \downarrow 0$, and (50) gives both $V(m) \rightarrow 0$ and $V(m) > 0$ for all sufficiently large finite m . Choose one such point. Endpoint vanishing makes the value outside some compact subinterval smaller than half its value at that point; continuity supplies a positive global maximizer on the compact interval. Equation (49) directly contradicts strict increase near zero when $\beta < 1/2$. No analogous derivative or optimizer-switch control has been proved for $\beta \geq 1/2$. $\square$

Proof of Proposition 3. The binomial identity in (57) follows by integrating $\mathbb{E}[t^X] = (1 - a + at)^{n-1}$. Given universal rejection, each rival cost is uniform on $(a, 1)$, giving θ_j , (58), and cutoff indifference (59). Conditional on posting, completion is

$$1 - (1 - a)^n + (1 - a)^n \sum_j \eta_j C_{j,n}.$$

Substitute (58) into the second term and use (59); the expression becomes $ns_n(a)[a + (p_1 - a)/\delta]$. Multiplication by posting mass $1 - p_1$ proves (60).

For an arbitrary nearby interior cutoff put $d = p - a$. Rearranging its indifference condition gives

$$\begin{aligned} [s_n(a) - \delta(1 - a)^{n-1}\alpha\{(\rho - \eta)s_n(\theta_1) + \eta s_n(\theta_2)\}]d \\ = \delta(1 - a)^{n-1}\alpha\eta s_n(\theta_2)\varepsilon. \end{aligned}$$

Because $s_n(a) \geq (1 - a)^{n-1}$, $s_n(\theta_j) \leq 1$, and $\eta \leq \rho$, every equilibrium satisfies

$$0 \leq d \leq \frac{\delta\alpha\rho}{1 - \delta\alpha\rho}\varepsilon.$$

The analogous boundary inequality excludes $a = 0$ for small ε ; at $a = p$ the cutoff type strictly prefers waiting. Thus all equilibria are interior and uniformly localize at p .

Write $a = p - \kappa\varepsilon + o(\varepsilon)$. Terminal coverage is first order in ε , the rider switch converges uniformly to p/β , and the rescue mass converges to $\rho(p)$. Divide cutoff indifference by ε and take the limit:

$$s_n(p)\kappa = \delta(1 - p)^{n-1}\alpha\rho(p)(1 + \kappa).$$

The denominator is positive because $s_n(p) \geq (1 - p)^{n-1}$ and $\delta\alpha\rho(p) < 1$, yielding (61). Differentiating (60) along the branch yields (62). The rescaled cutoff margin converges in C^1 on the compact localization interval to the displayed affine equation, whose slope is the positive denominator in (61). Existence from the general cutoff argument and this one-crossing limit give local uniqueness. Finally, $s_n(a) = n^{-1} \sum_{r=0}^{n-1} (1 - a)^r$ is strictly decreasing iff $n \geq 2$, which proves the sign statement.

□